

Coming out from the shadows: facultative slave-making ants reveal their chemical identity during colony development

Behavioral Ecology and Sociobiology

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1 Introduction

This document presents graphical results generated as output from analytic workflows written in *R* language and available in the public GitHub repository - <https://github.com/TomVuod/sangchem/tree/supp>.

2 Changes in CHC amount over time

This section contains a series of linear (mixed) models fitted to examine how the proportion of *F. sanguinea* workers in a colony is associated with the amount of CHC, analyzed as a whole or in subsets. Included are summary reports, results of the Shapiro-Wilk test of normality of conditional residuals, as well as diagnostic plots and tests generated with the use of **DHARMA** R package[Hartig, 2024]. The response variable often needed to be transformed to meet model assumptions. Random terms with no variance have been dropped.

In model specification (1|random_factor) denotes random intercept (separate for each level of the random factor) and ((1|random_factor_1:random_factor_2)) denotes random intercepts generated from the combinations of two factors. β_0 and ϵ denote intercept and error term, respectively.

2.1 Colony growth and body size

In the analyses below, normalized CHC amounts were used as the response variable. Because body shape in both species deviates from an isometric growth pattern (see Section 7), normalizing CHC amounts by the square of head width may introduce bias. To address this, we first examined the relationship between body size (head width) and the proportion of *F. sanguinea* workers in a colony, and then incorporated the square of head width as an explanatory variable in the linear model.

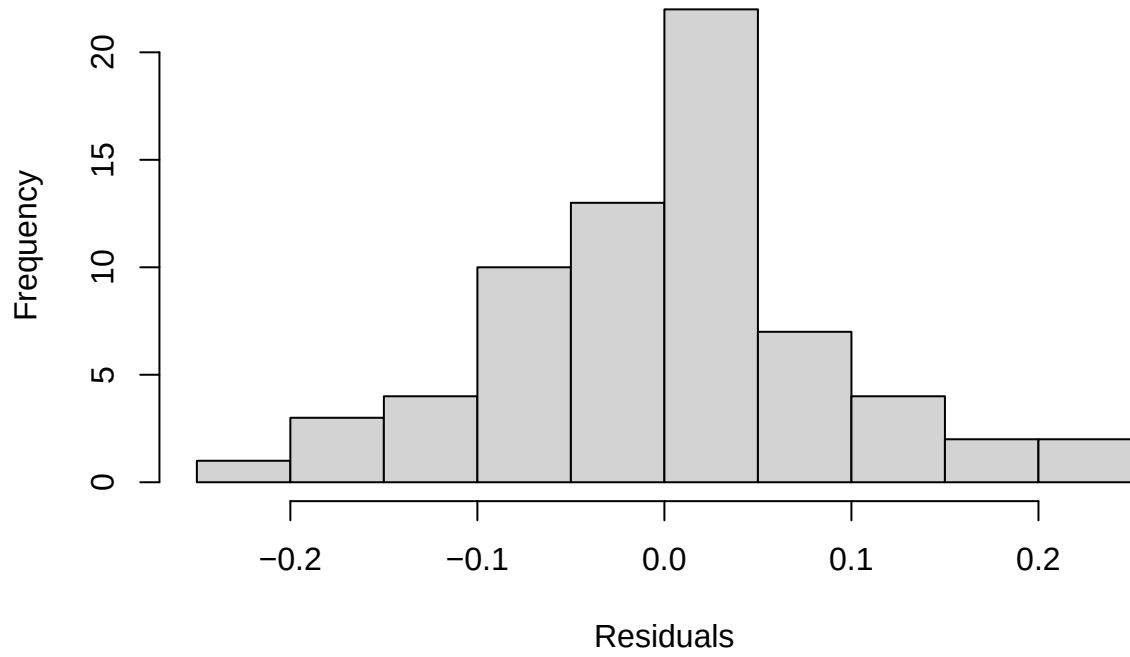
$$\text{head_width} = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + \epsilon,$$

```

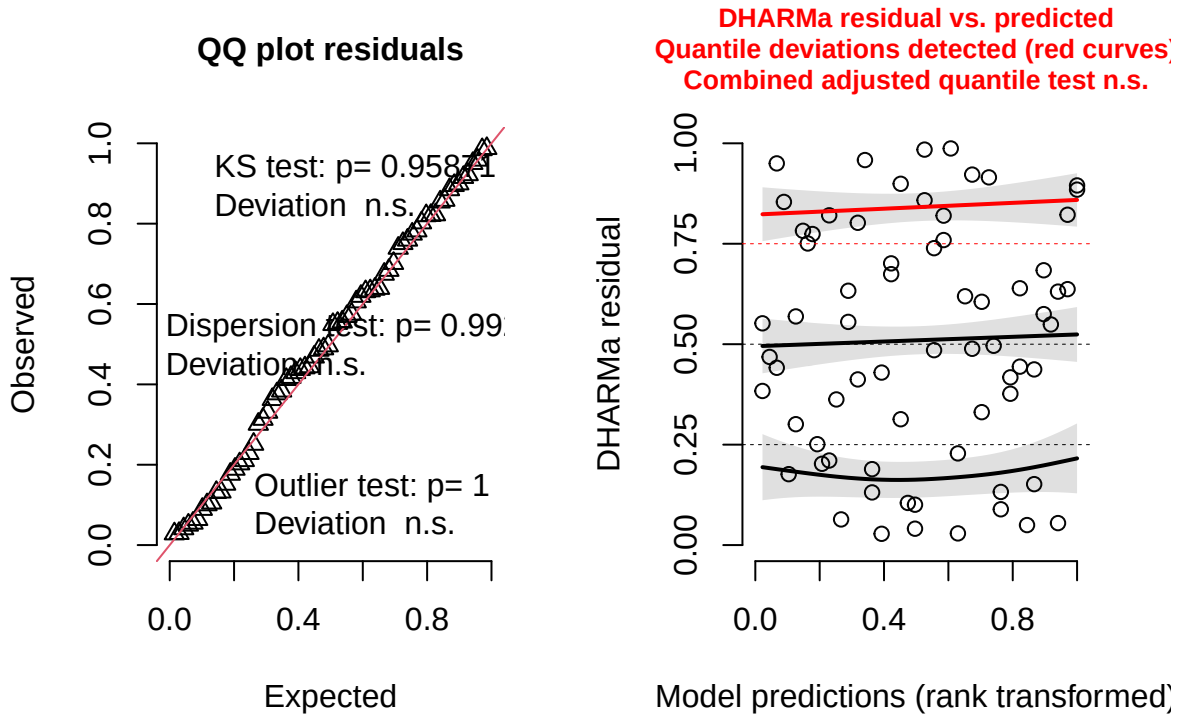
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: head_width ~ sang_prop + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: -103.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.10138 -0.61076  0.06058  0.43106  2.28980
##
## Random effects:
## Groups Name Variance Std.Dev.
## colony (Intercept) 0.003155 0.05617
## Residual 0.009110 0.09545
## Number of obs: 68, groups: colony, 16
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  1.21806    0.02641 38.92702  46.121 < 2e-16 ***
## sang_prop    0.12514    0.04344 62.09975   2.881  0.00544 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## sang_prop -0.715
##
##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.9861, p-value = 0.6514

```

Model conditional residuals



DHARMa residual



2.2 Change in the total CHC mass

In this series of models the total CHC mass (normalized by assuming head width of 1.3 mm) was regressed on the proportion of *Formica sanguinea* workers in the colony.

2.2.1 Mature *F. sanguinea*

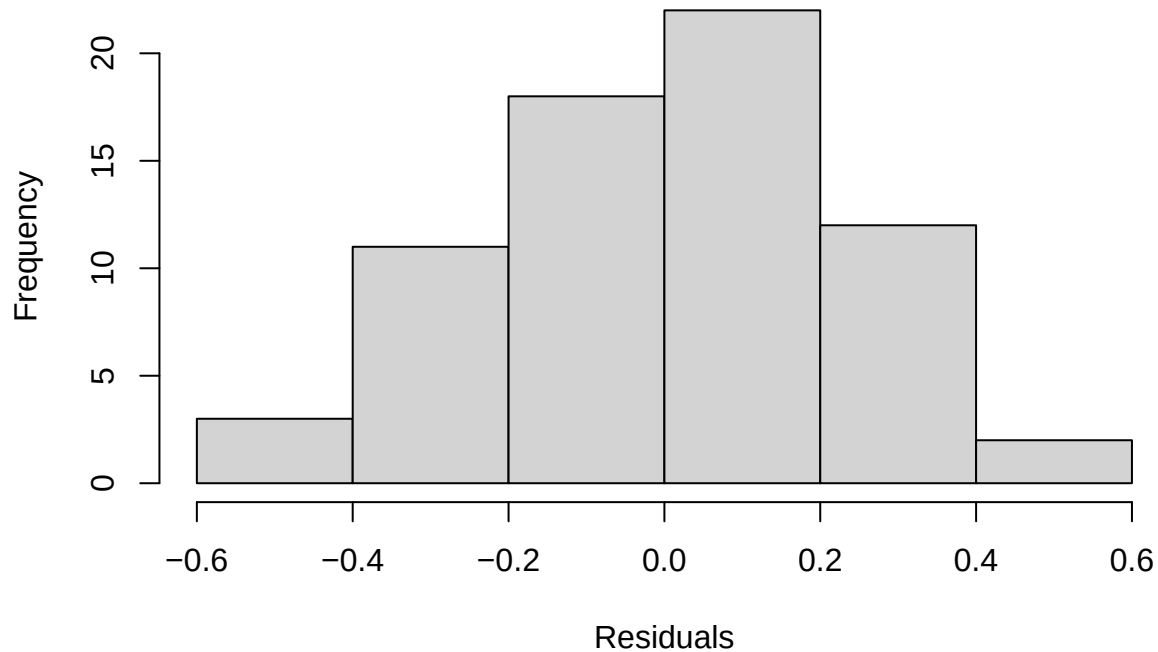
$\log(\text{CHC_mass_sanguinea}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID}:\text{sampling_occasion_ID}) + \epsilon$,
where CHC_mass_sanguinea denotes the total normalized CHC mass on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass)) ~ sang_prop + (1 | colony) + (1 | colony:census_date)
## Data: model_input
##
## REML criterion at convergence: 92.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.83723 -0.39249  0.01222  0.53571  1.73459
##
## Random effects:
## Groups              Name             Variance Std.Dev.
## colony:census_date (Intercept) 0.12975  0.3602
```

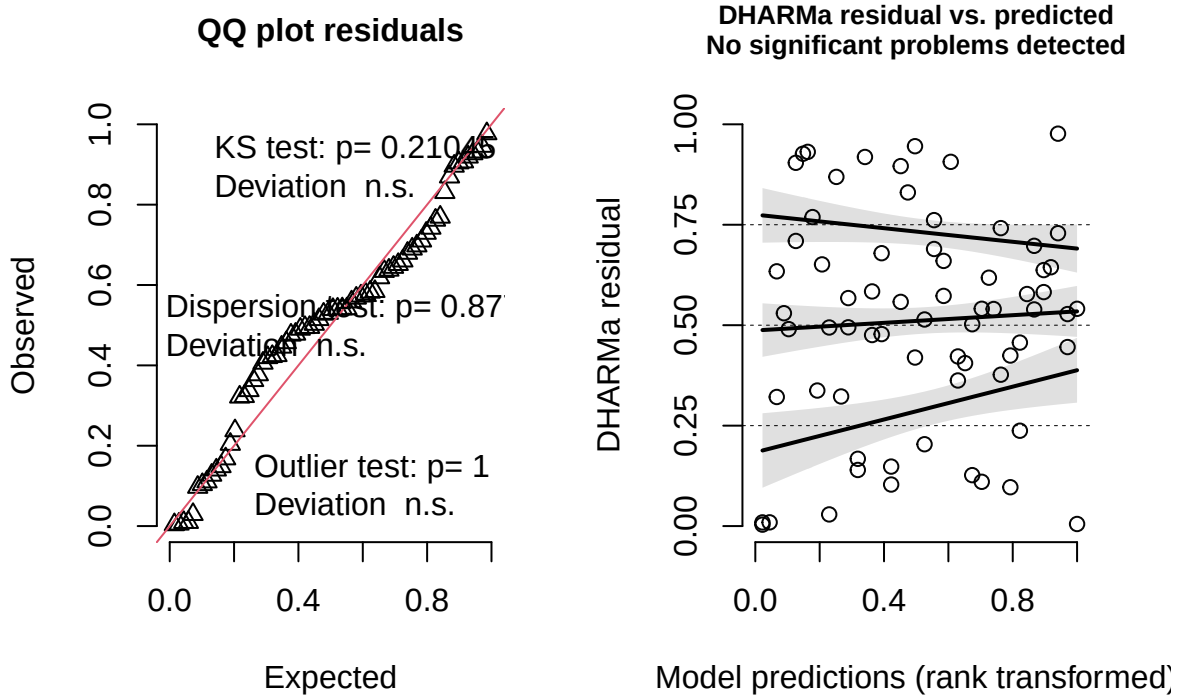
```
## colony (Intercept) 0.02626 0.1620
## Residual 0.10333 0.3215
## Number of obs: 68, groups: colony:census_date, 44; colony, 16
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 1.2035 0.1295 36.0905 9.296 4.11e-11 ***
## sang_prop 0.7991 0.2392 36.9985 3.340 0.00192 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## sang_prop -0.786

##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.98623, p-value = 0.6591
```

Model conditional residuals



DHARMa residual



The model shows no significant effect of body size on CHC amount in *F. sanguinea*, and this term will therefore be excluded from subsequent models for this species.

2.2.2 *F. fusca*

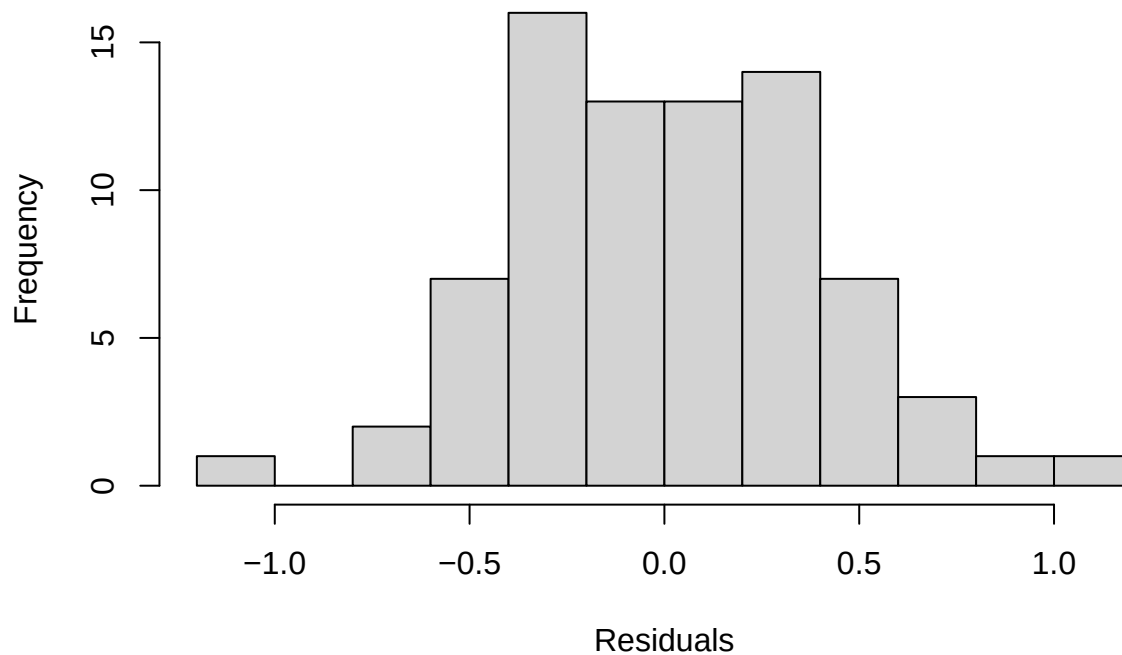
$$\log(\text{CHC_mass_fusca}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + \epsilon,$$

where CHC_mass_fusca denotes the total normalized CHC mass on the body of adult *F. fusca* worker.

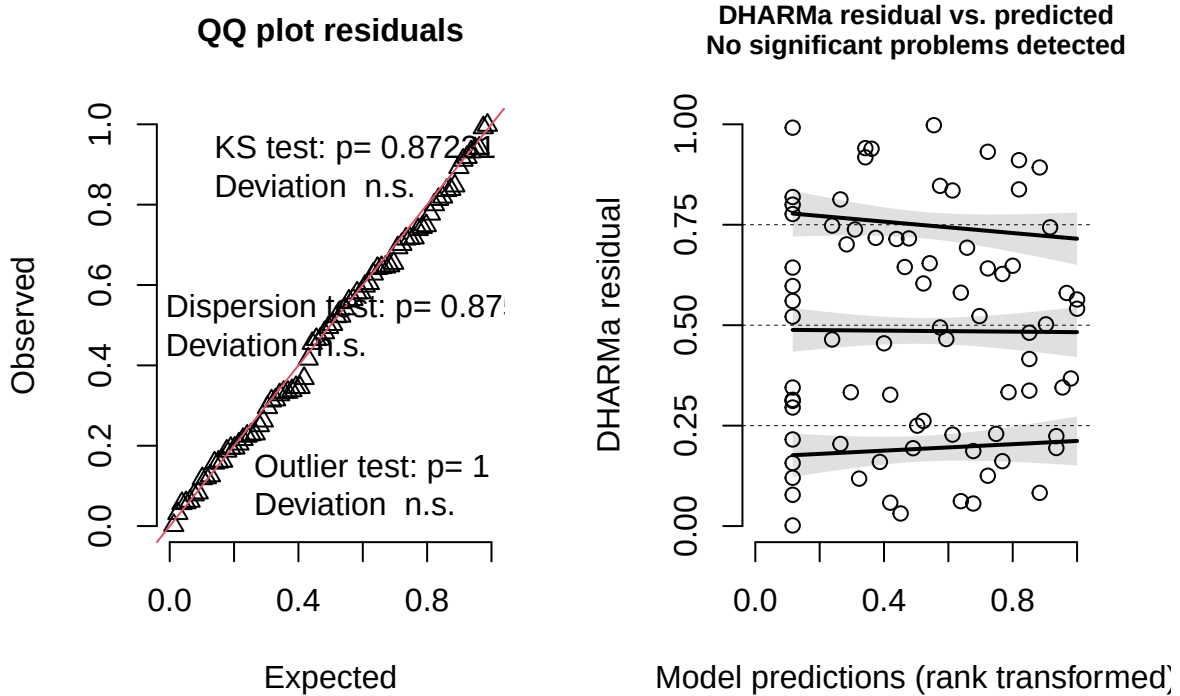
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass)) ~ sang_prop + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: 103.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.79685 -0.62513 -0.00685  0.63754  2.49220
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## colony   (Intercept)  0.05014   0.2239
## Residual                    0.17722   0.4210
## Number of obs: 78, groups: colony, 20
##
```

```
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  0.81635    0.08805 31.60123   9.272 1.57e-10 ***
## sang_prop    0.17928    0.17517 70.00046   1.024    0.31
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## sang_prop -0.594
##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99196, p-value = 0.9107
```

Model conditional residuals



DHARMa residual



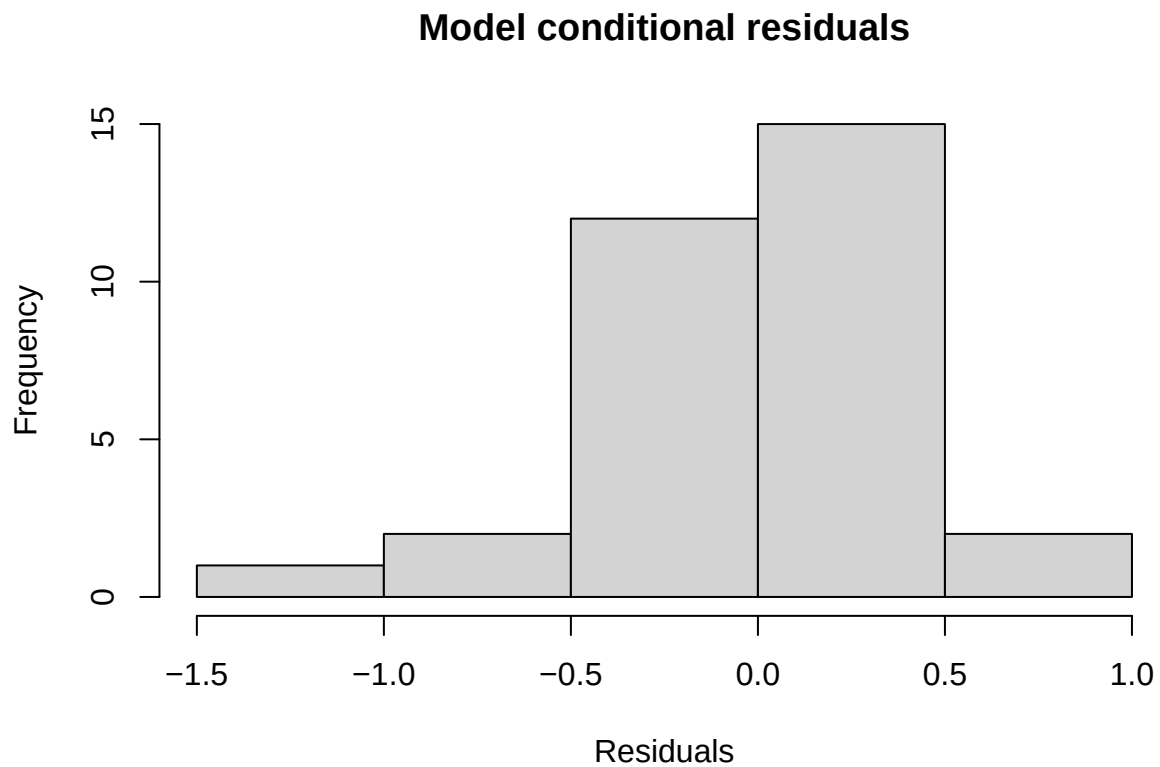
2.2.3 Callow *F. sanguinea*

$\log(\text{CHC_mass_callow}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID}:\text{sampling_occasion_ID}) + \epsilon$,
where CHC_mass_callow denotes the total normalized CHC mass on the body of callow *F. sanguinea* worker.

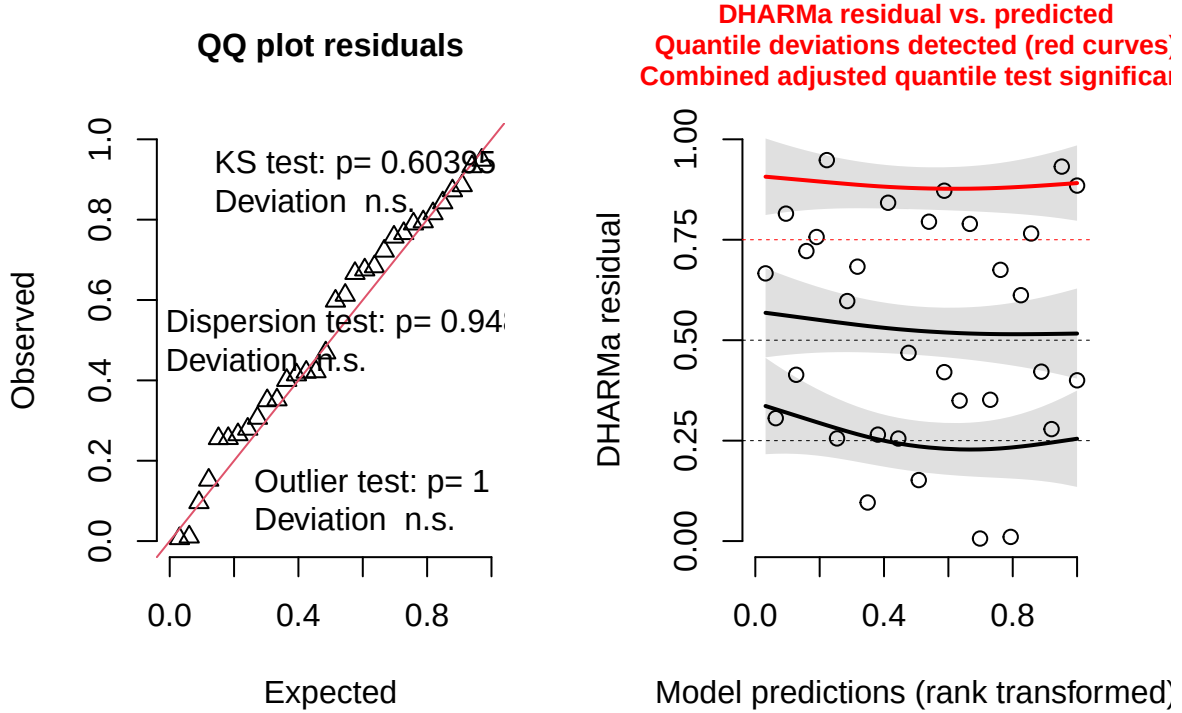
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass)) ~ sang_prop + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: 51.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.16539 -0.61246  0.07593  0.62418  1.49264
##
## Random effects:
## Groups Name Variance Std.Dev.
## colony (Intercept) 0.06331 0.2516
## Residual 0.22509 0.4744
## Number of obs: 32, groups: colony, 16
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
```

```
## (Intercept)  0.03373    0.17027 28.88675    0.198  0.84435
## sang_prop    1.05723    0.30345 25.91578    3.484  0.00177 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## sang_prop -0.775

##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.95692, p-value = 0.2258
```



DHARMA residual



Similarly to mature *F. sanguinea* ants, in callow workers there is negative but not significant relationship between body size and normalized CHC amount.

2.3 Change of the CHC characteristic of *F. sanguinea*

In this section the subset of peaks were used to calculate normalized CHC mass. These were peaks identified as *F. sanguinea* markers.

2.3.1 Mature *F. sanguinea* ants

$$\log(\text{CHC_mass_mature}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID:sampling_occasion_ID}) + \epsilon,$$

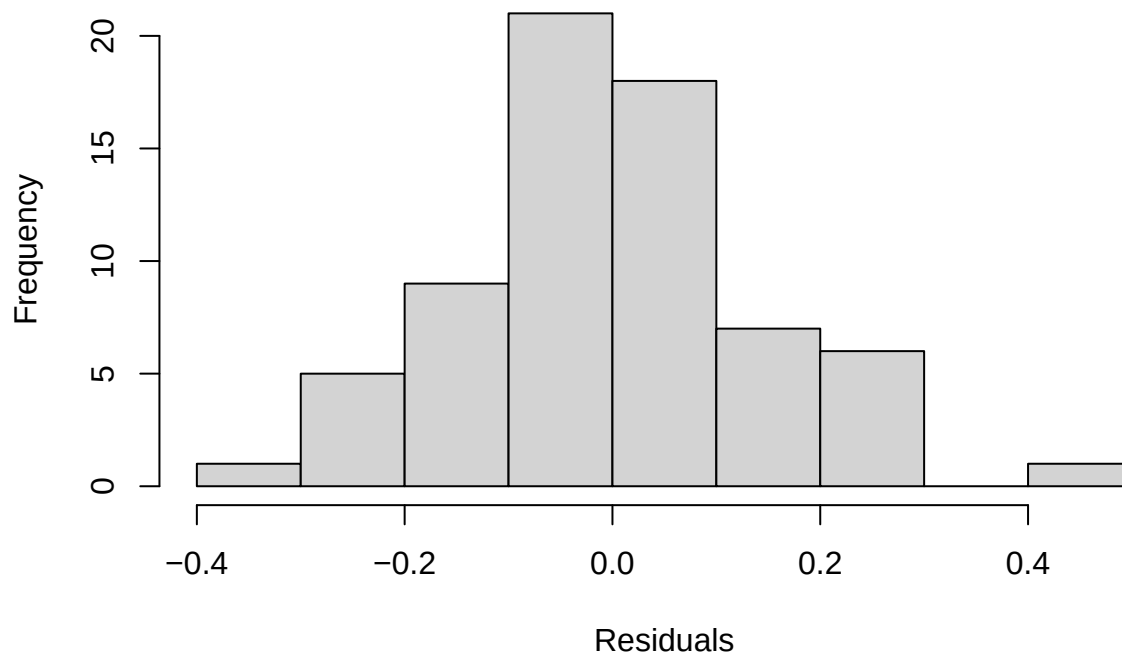
where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass + 1)) ~ sang_prop + (1 | colony:census_date) + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: 30.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8821 -0.3658 -0.0362  0.3439  2.1883
```

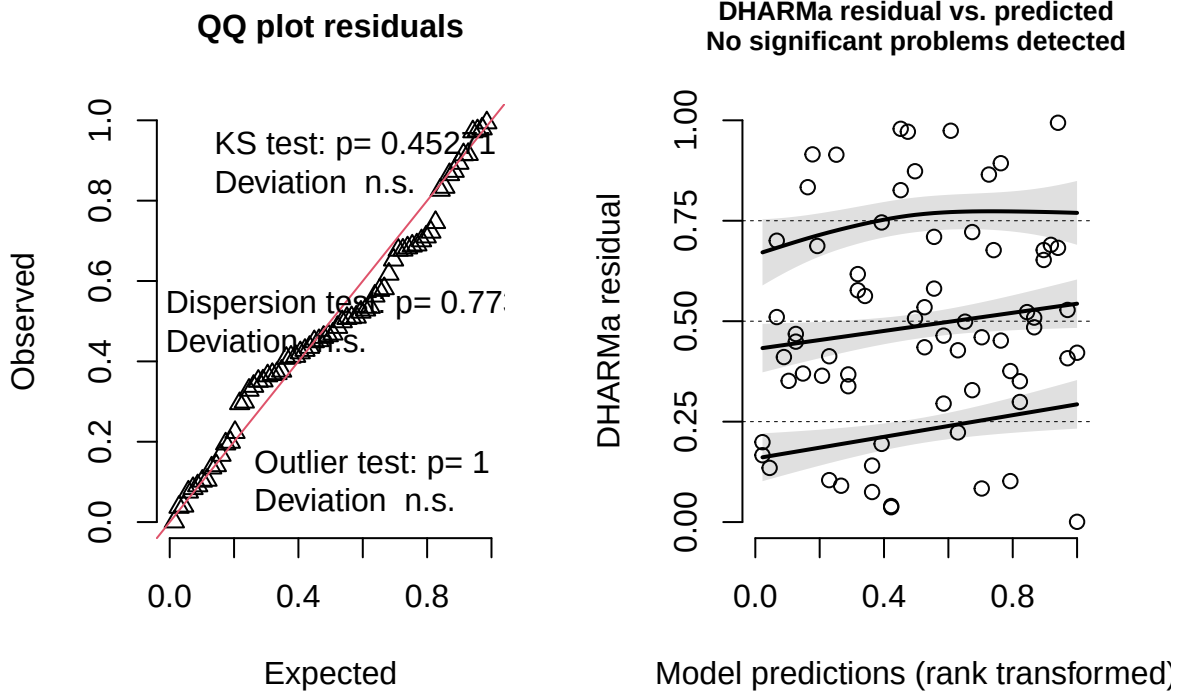
```
##
## Random effects:
##   Groups           Name          Variance Std.Dev.
## colony:census_date (Intercept) 0.06558  0.2561
## colony              (Intercept) 0.00000  0.0000
## Residual                        0.03804  0.1950
## Number of obs: 68, groups: colony:census_date, 44; colony, 16
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   0.2833    0.0805  40.1068   3.519  0.00109 **
## sang_prop     1.3131    0.1564  39.5702   8.394 2.54e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## sang_prop -0.822
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.98702, p-value = 0.7046
```

Model conditional residuals



DHARMA residual



2.3.2 *F. fusca* ants

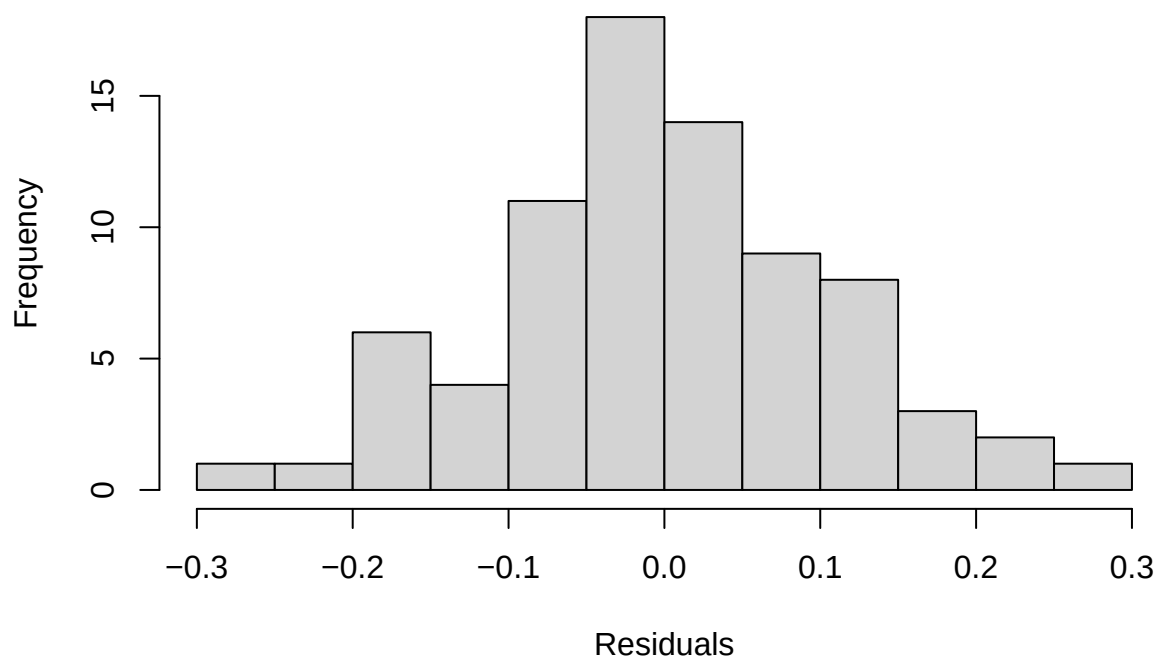
$\sqrt{(\text{CHC_mass_fusca})} = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID:sampling_occasion_ID}) + \epsilon$,
where CHC_mass_fusca denotes the normalized mass of *F. sanguinea* markers on the body of adult *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(normalized_mass)) ~ sang_prop + (1 | colony:census_date)
## Data: model_input
##
## REML criterion at convergence: -58.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.06253 -0.51814 -0.07238  0.45433  1.95590
##
## Random effects:
## Groups              Name            Variance Std.Dev.
## colony:census_date (Intercept) 0.009805 0.09902
## Residual                0.016560 0.12869
## Number of obs: 78, groups: colony:census_date, 55
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
```

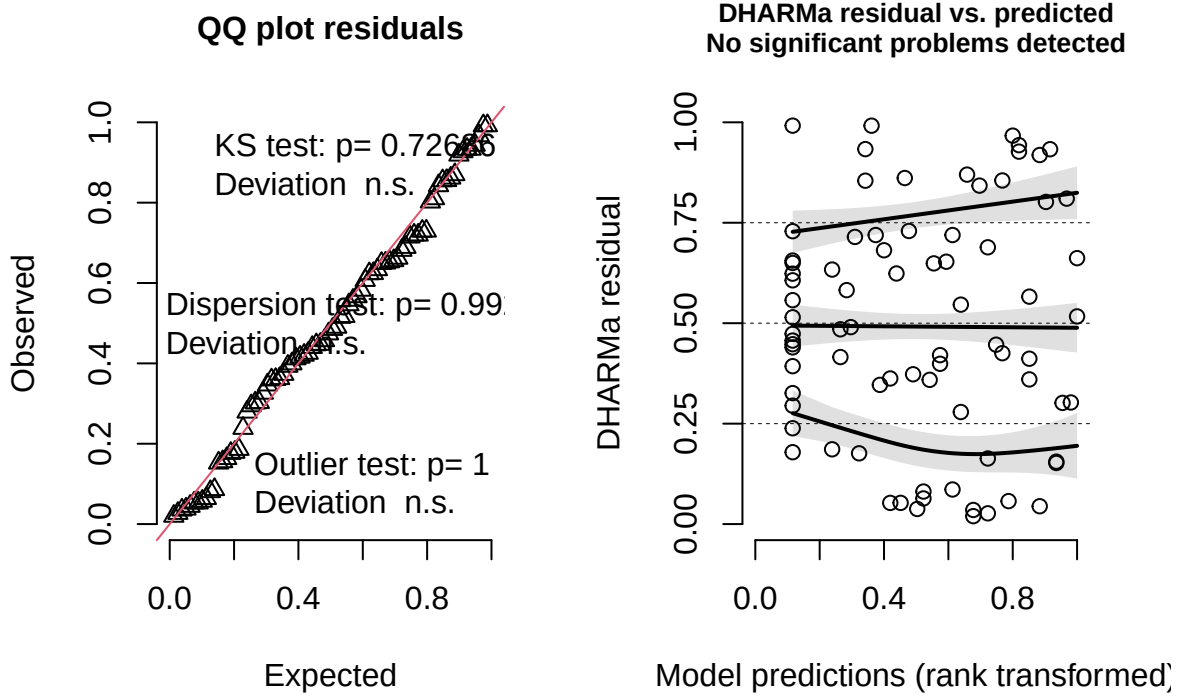
```
## (Intercept)  0.41024    0.02938 54.47670  13.965 < 2e-16 ***
## sang_prop   0.61852    0.06935 51.99602   8.919 4.63e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## sang_prop -0.729

##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99067, p-value = 0.8465
```

Model conditional residuals



DHARMa residual



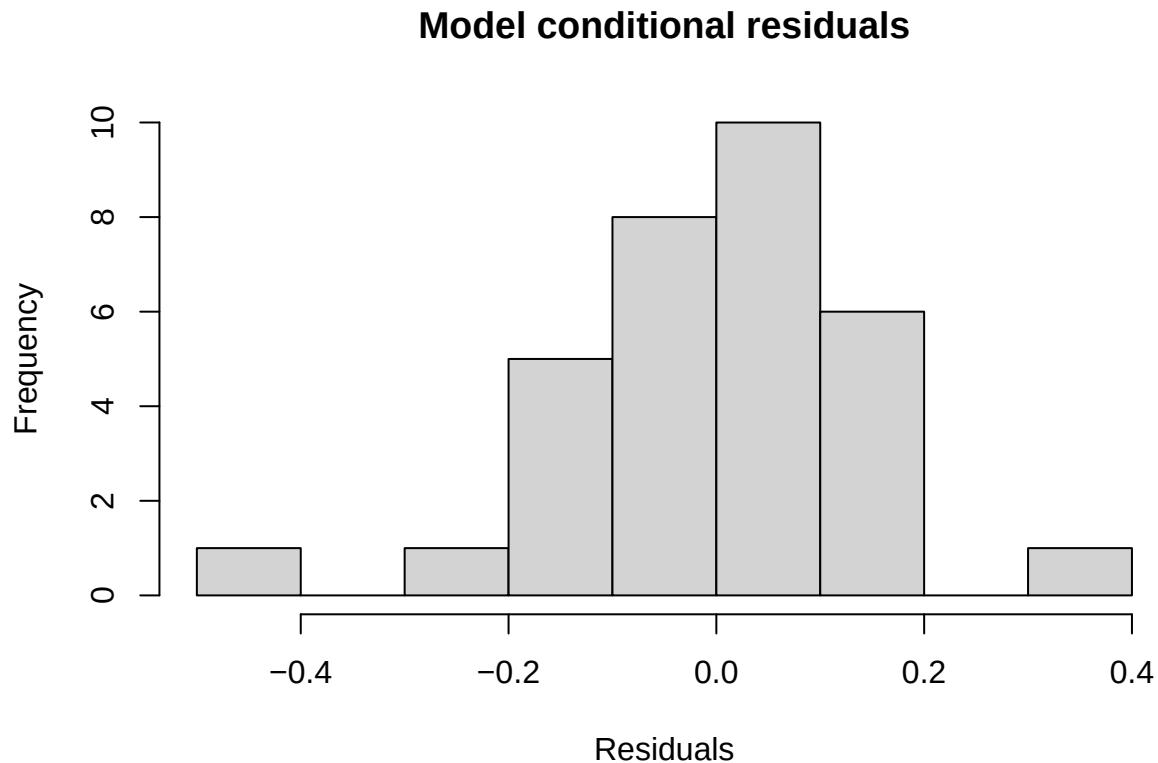
2.3.3 Callow *F. sanguinea* ants

$$\sqrt[3]{(\text{CHC_mass_mature})} = \beta_0 + \text{sanguinea_proportion} + \epsilon,$$

where CHC_mass_mature denotes the normalized mass of *F. sanguinea* markers on the body of callow *F. sanguinea* worker.

```
##
## Call:
## lm(formula = I((normalized_mass)^(1/3)) ~ sang_prop, data = model_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41432 -0.06803  0.02430  0.07590  0.35296
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.31592    0.04638   6.812 1.48e-07 ***
## sang_prop     0.71797    0.08719   8.235 3.42e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1473 on 30 degrees of freedom
## Multiple R-squared:  0.6933, Adjusted R-squared:  0.6831
## F-statistic: 67.81 on 1 and 30 DF, p-value: 3.423e-09
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.96738, p-value = 0.4305
```



2.4 Change of the CHC characteristic of *F. fusca*

Similarly to the analysis with the use of markers of *F. sanguinea* workers, the procedure was repeated using the mass of *F. fusca* markers as a response variable.

2.4.1 Mature *F. sanguinea* ants

$\log(\text{CHC_mass_mature}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID:sampling_occasion_ID}) + \epsilon$,
 where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of adult *F. sanguinea* worker.

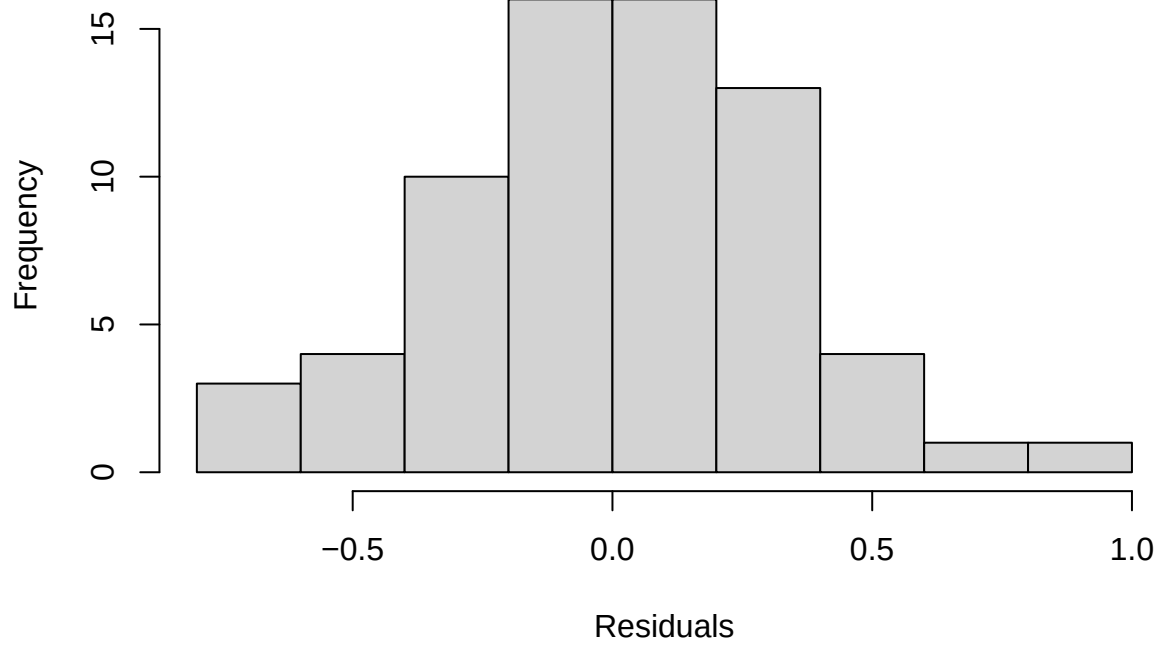
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass)) ~ sang_prop + (1 | colony:census_date) + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: 131.2
##
```

```

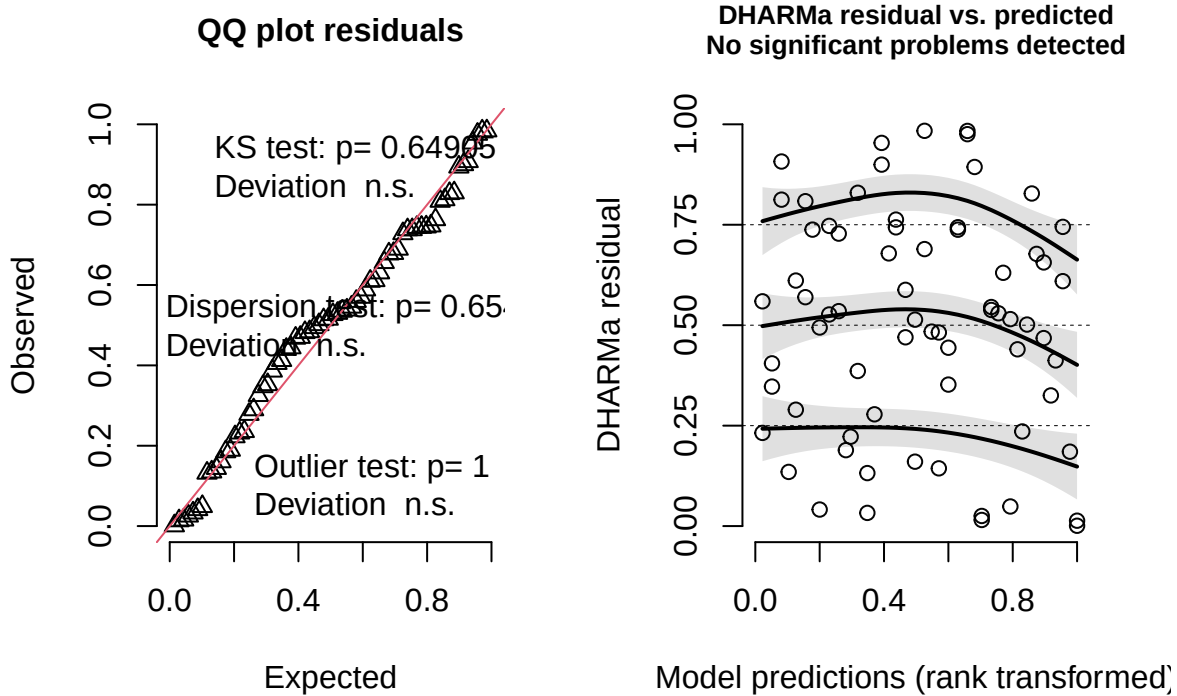
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.82877 -0.45873  0.05316  0.55616  1.98511
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## colony:census_date (Intercept) 0.1471    0.3835
## colony              (Intercept) 0.2359    0.4857
## Residual                        0.1804    0.4247
## Number of obs: 68, groups: colony:census_date, 44; colony, 16
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.4566      0.1947  33.8202  -2.345 0.025027 *
## sang_prop    -1.1016      0.2951  33.2726  -3.733 0.000707 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## sang_prop -0.656
##
##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.98971, p-value = 0.8512

```

Model conditional residuals



DHARMA residual



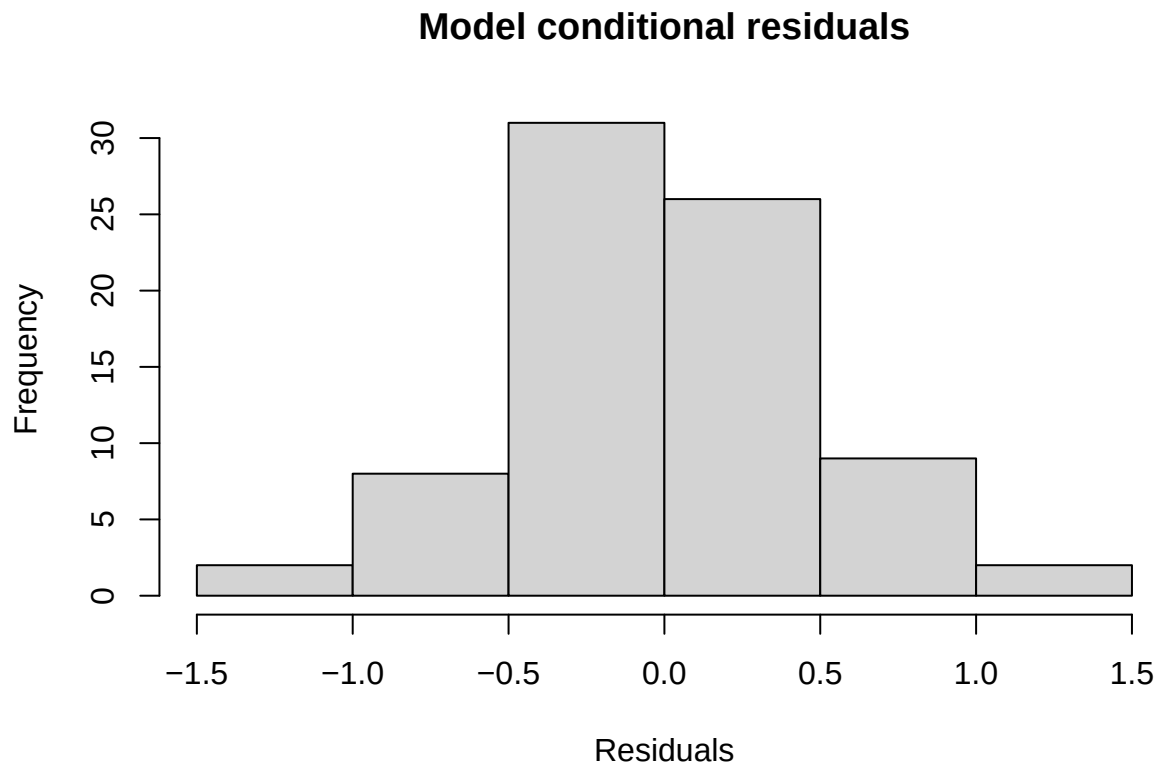
2.4.2 *F. fusca* ants

$\log(\text{CHC_mass_mature} + 0.01) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + (1|\text{colony_ID}:\text{sampling_occasion_ID}) + \epsilon$,
where CHC_mass_mature denotes the normalized mass of *F. fusca* markers on the body of *F. fusca* worker.

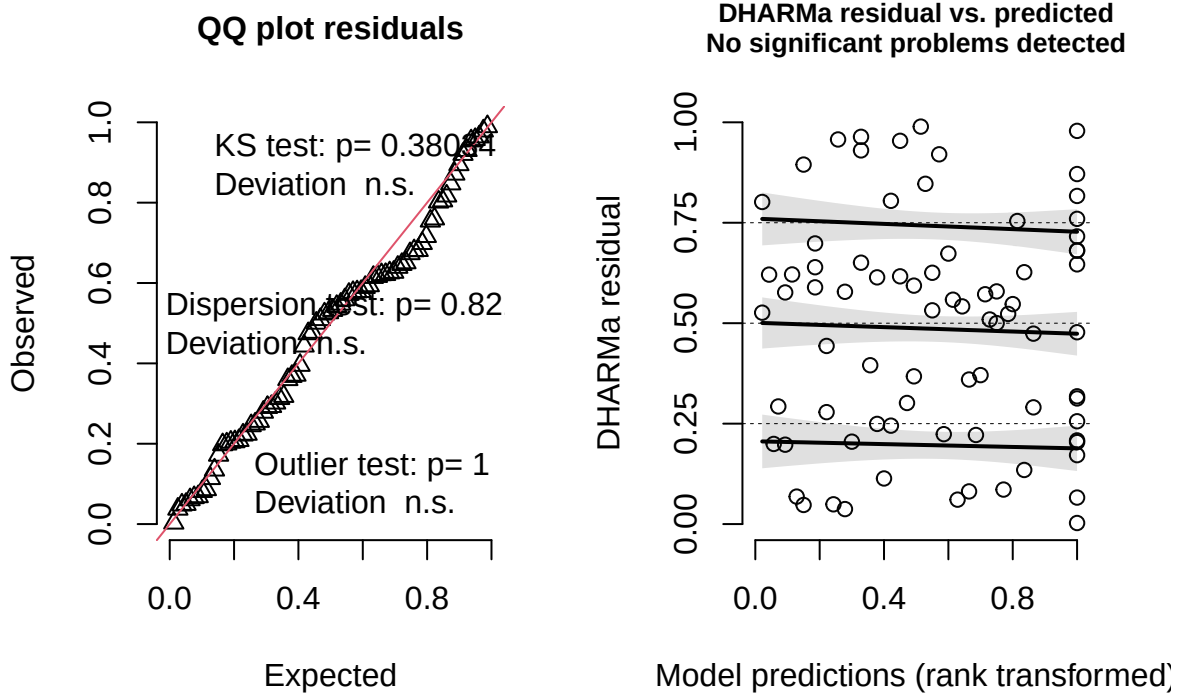
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass + 0.01)) ~ sang_prop + (1 | colony) + (1 | colony:census_date)
## Data: model_input
##
## REML criterion at convergence: 159.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.34120 -0.60468 -0.05522  0.59876  2.14890
##
## Random effects:
## Groups              Name             Variance Std.Dev.
## colony:census_date (Intercept)  0.0000    0.0000
## colony              (Intercept)  0.2325    0.4821
## Residual                        0.3213    0.5668
## Number of obs: 78, groups: colony:census_date, 55; colony, 20
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
```

```
## (Intercept)  -0.8547      0.1462 27.2550  -5.844 3.08e-06 ***
## sang_prop    -0.8184      0.2436 65.7807  -3.360 0.0013 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## sang_prop -0.479
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.99138, p-value = 0.8835
```



DHARMA residual



2.4.3 Callow *F. sanguinea* ants

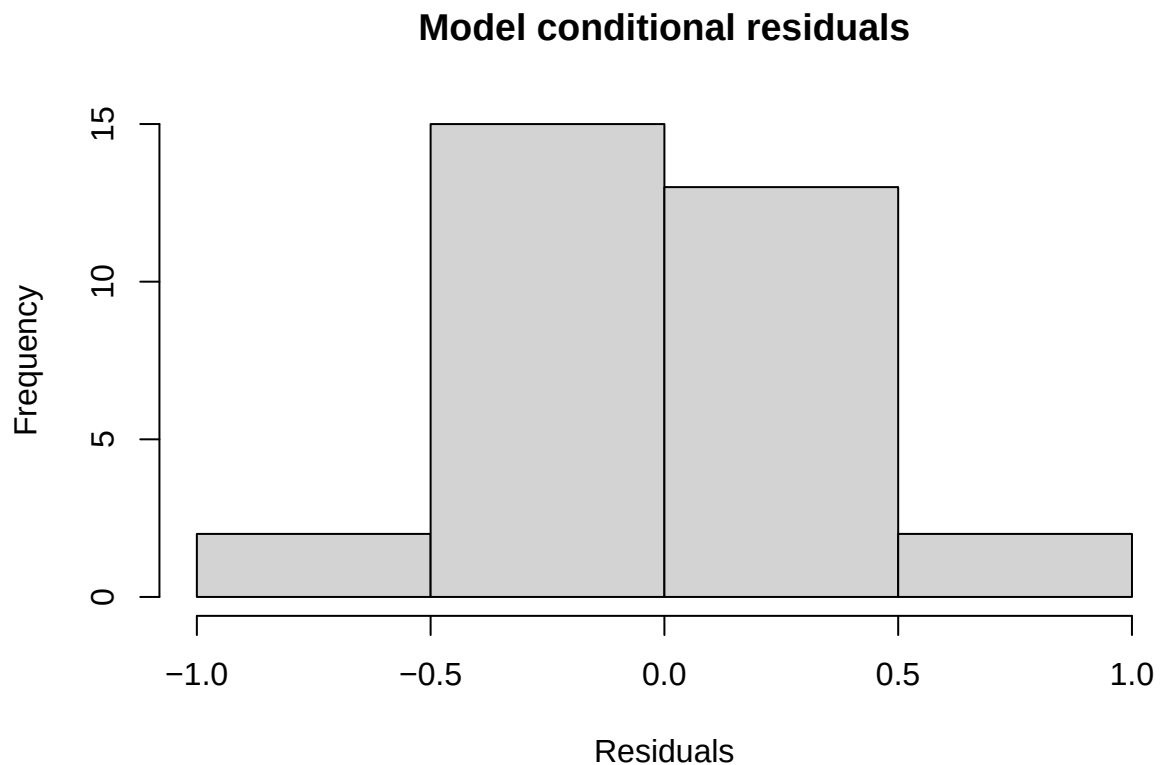
$$\log(\text{CHC_mass_mature}) = \beta_0 + \text{sanguinea_proportion} + (1|\text{colony_ID}) + \epsilon,$$

where $\log(\text{CHC_mass_mature})$ denotes the normalized mass of *F. fusca* markers on the body of callow *F. sanguinea* worker.

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(log(normalized_mass)) ~ sang_prop + (1 | colony)
## Data: model_input
##
## REML criterion at convergence: 59.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.88270 -0.59925 -0.01403  0.65154  1.88156
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## colony   (Intercept) 0.2208   0.4699
## Residual                0.2225   0.4717
## Number of obs: 32, groups: colony, 16
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
```

```
## (Intercept) -1.66659    0.20293 28.05257  -8.213 6.04e-09 ***
## sang_prop   -0.07903    0.32299 19.69963  -0.245    0.809
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr)
## sang_prop -0.685

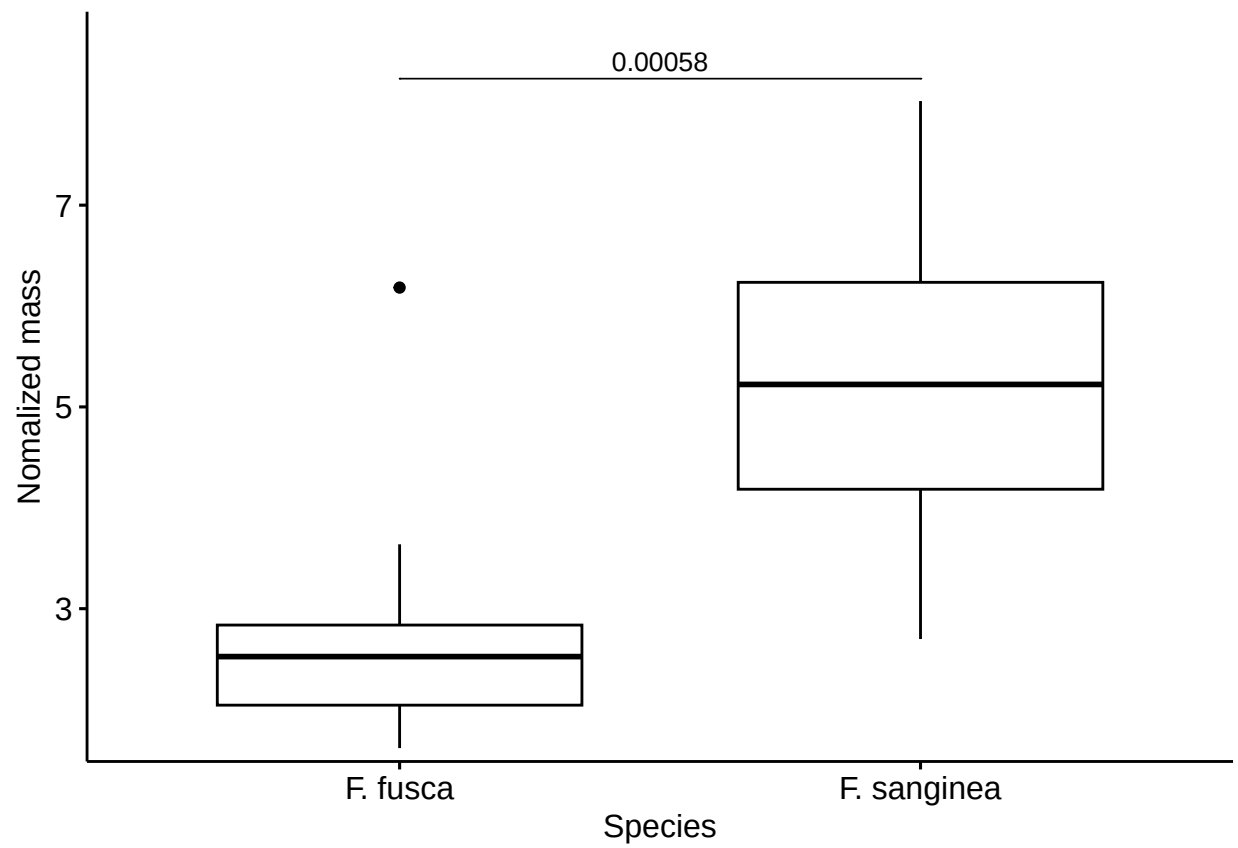
##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.98634, p-value = 0.9487
```



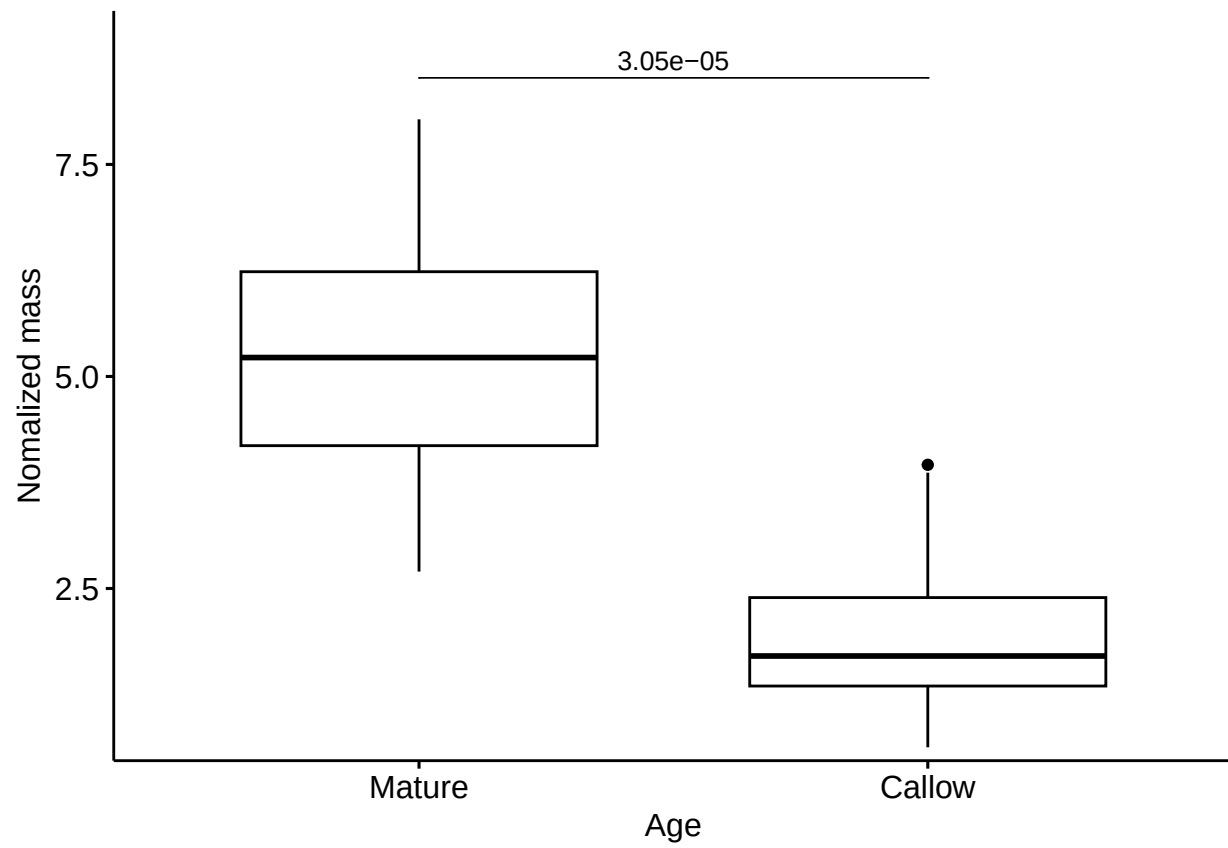
3 Comparison of the amount and proportions of CHC between species and age categories

This section presents the results of Wilcoxon tests comparing features of CHC profiles between callow and mature *F. sanguinea* ants as well as *F. fusca* slaves. Samples from the same colony were averaged before calculation of the final statistics to account for their non-independence.

3.1 Difference in the total CHC amount between *F. sanguinea* and *F. fusca*



3.2 Difference in the total CHC amount between callow and mature *F. sanguinea*



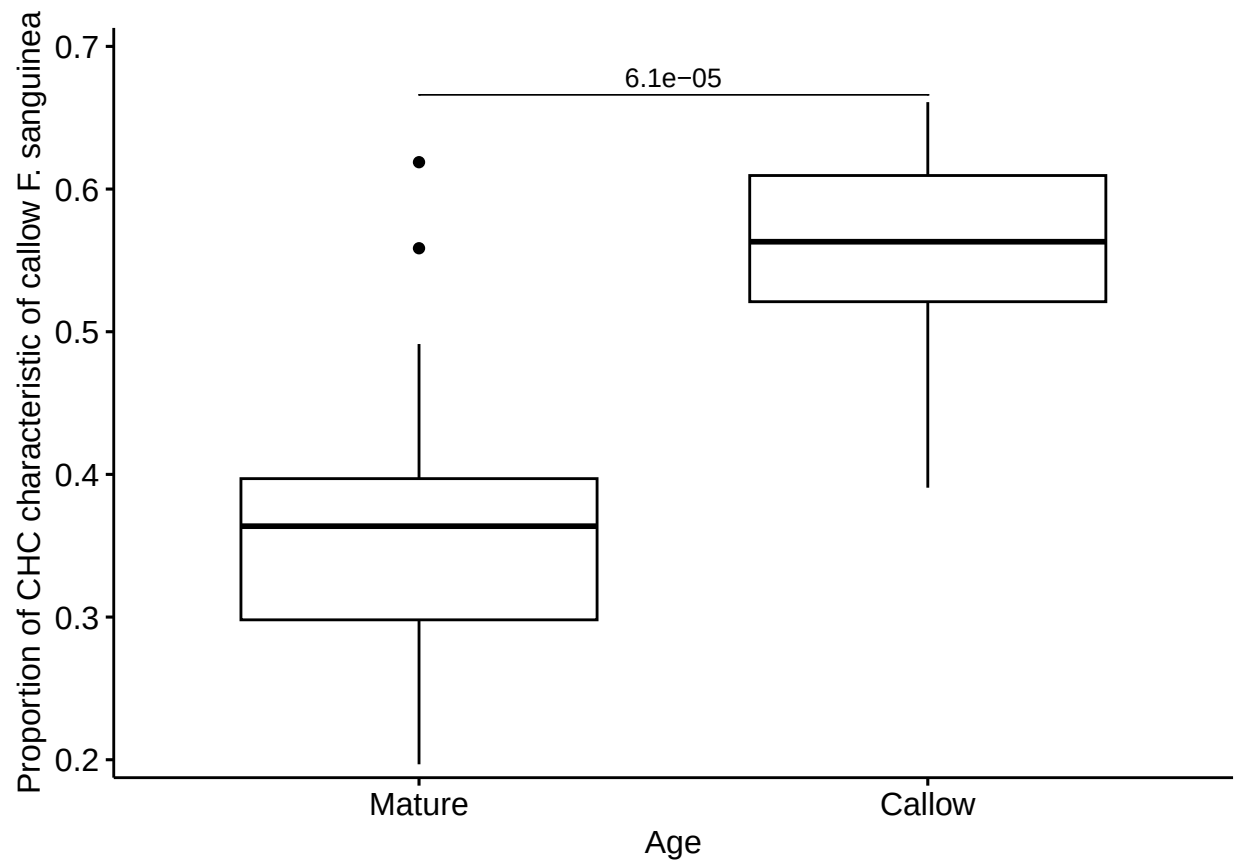
3.3 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature *F. sanguinea* and *F. fusca*



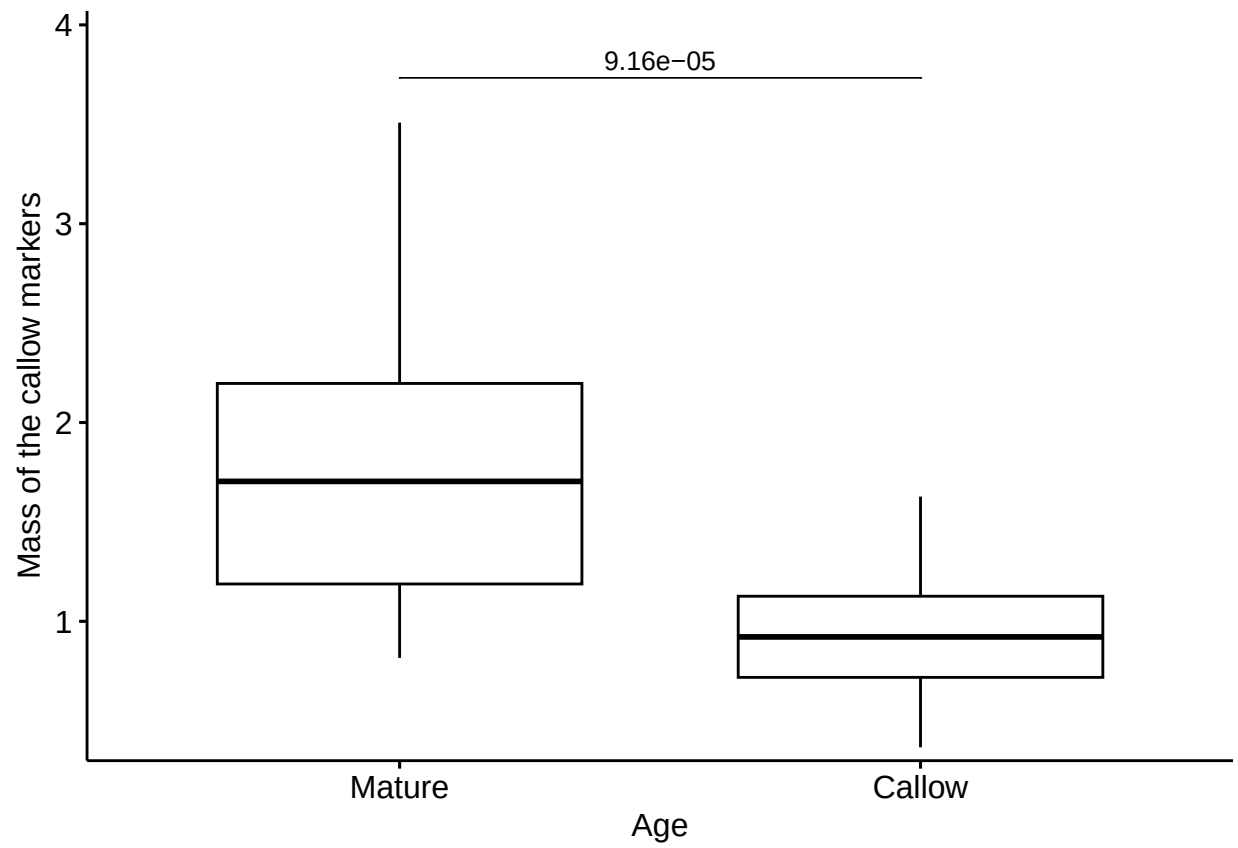
3.4 Difference in the proportion of CHC characteristic of *F. sanguinea* between mature and callow *F. sanguinea*



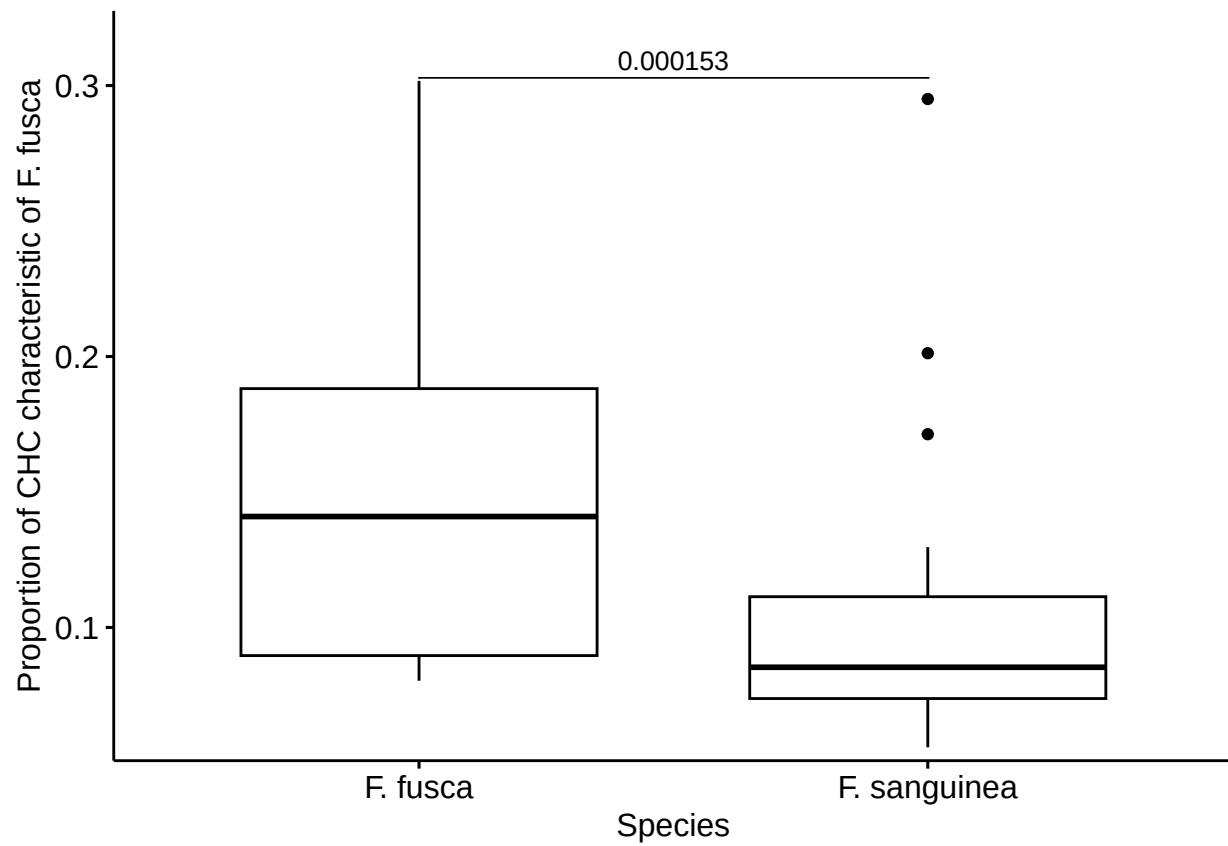
3.5 Difference in the proportion of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



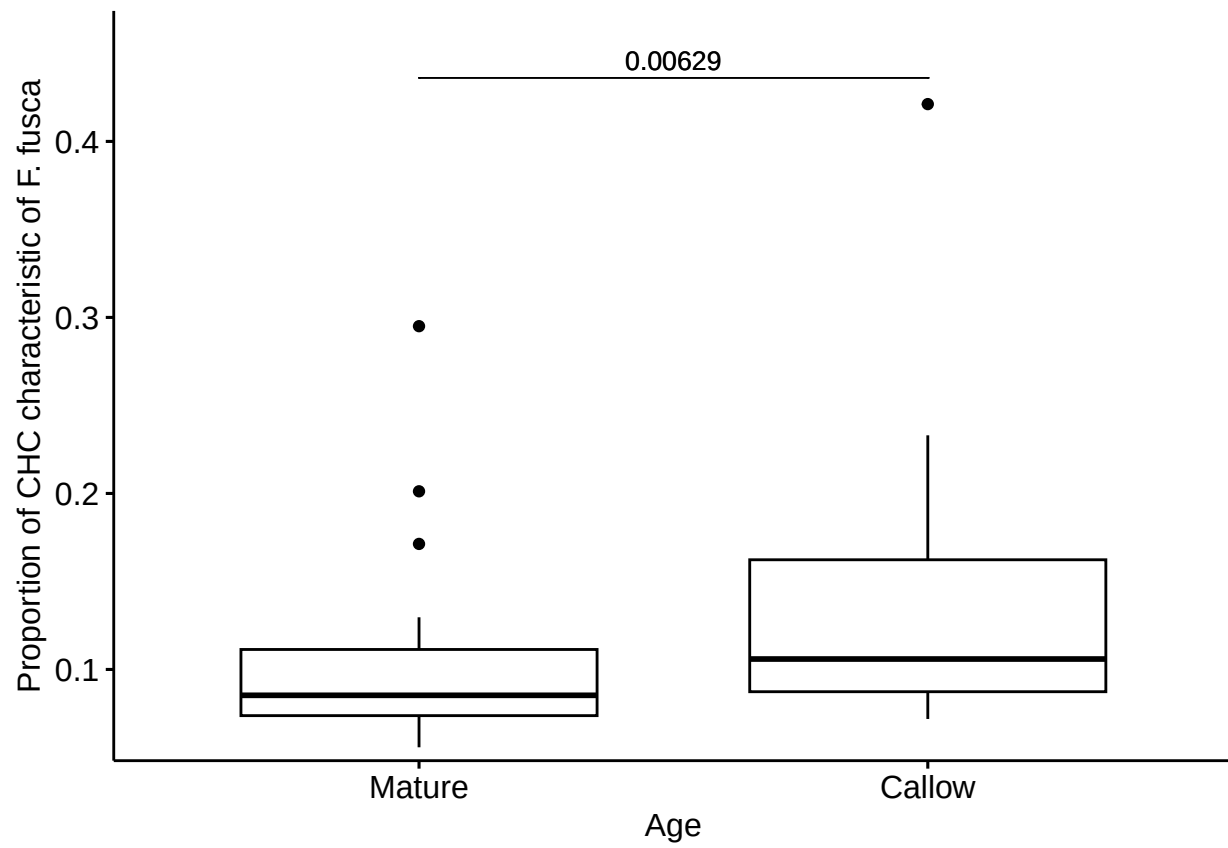
3.6 Difference in the amount of CHC characteristic of callow *F. sanguinea* between mature and callow *F. sanguinea*



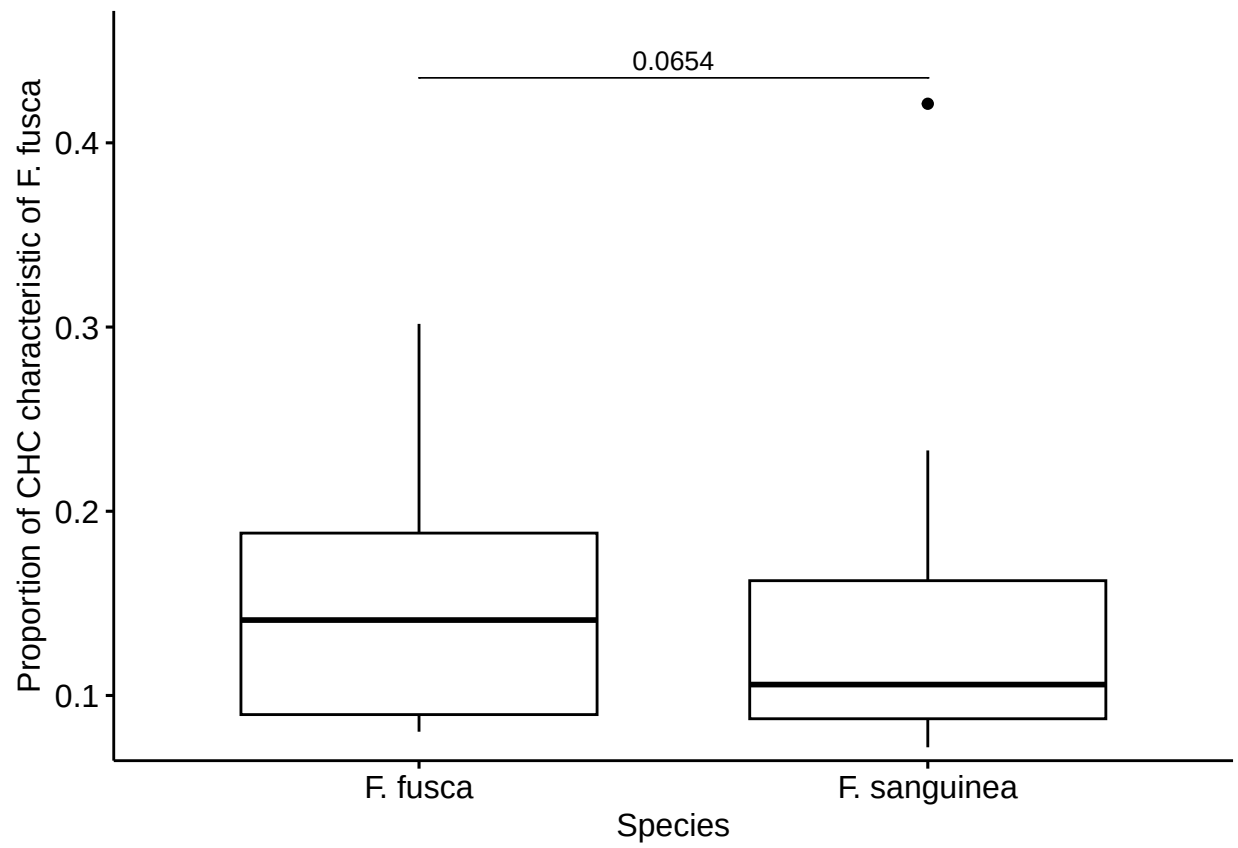
3.7 Difference in the proportion of CHC characteristic of *F. fusca* between mature *F. sanguinea* and *F. fusca*



3.8 Difference in the proportion of CHC characteristic of *F. fusca* between mature and callow *F. sanguinea*



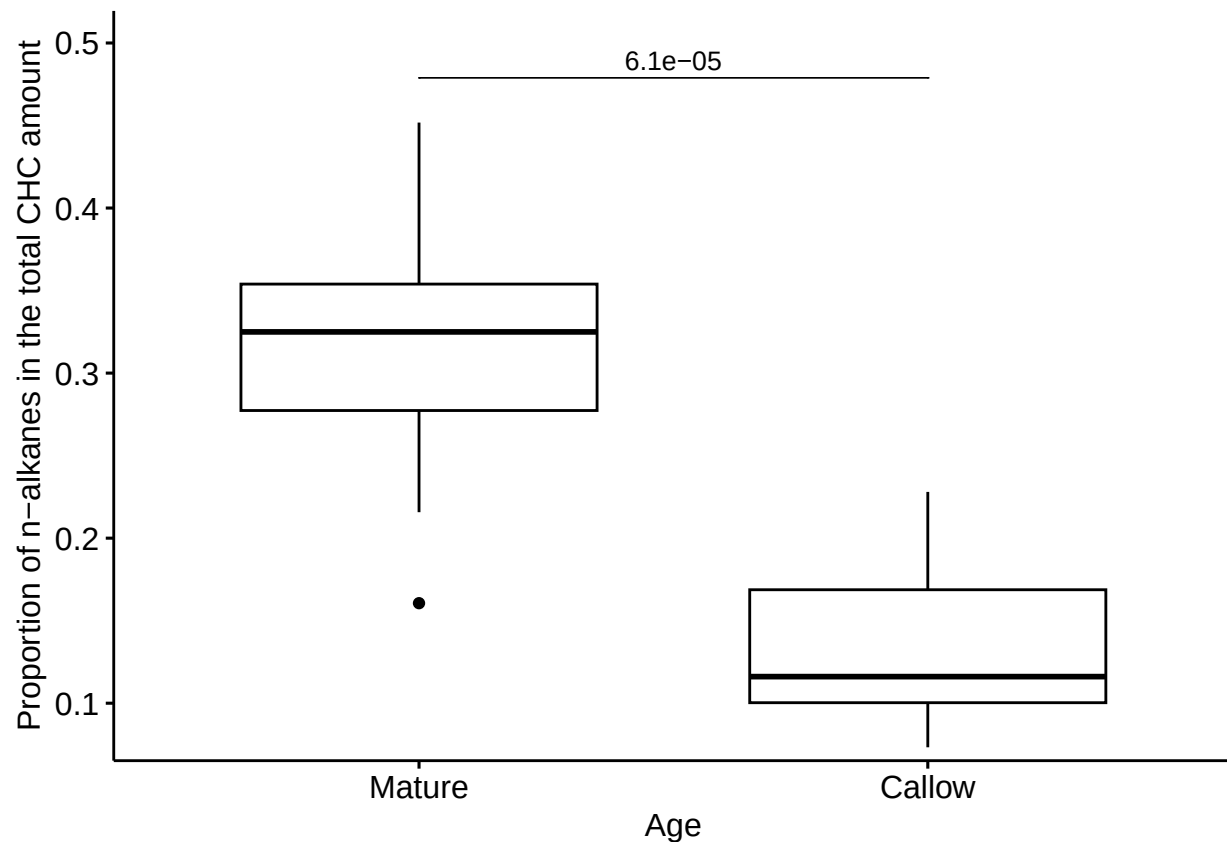
3.9 Difference in the proportion of CHC characteristic of *F. fusca* between
callow *F. sanguinea* and *F. fusca*



3.10 Difference in the mass of *n*-alkanes between callow and mature *F. sanguinea*



3.11 Difference in the proportion of *n*-alkanes between callow and mature *F. sanguinea*



4 Change in the CHC profile of separated callow *F. sanguinea* ants

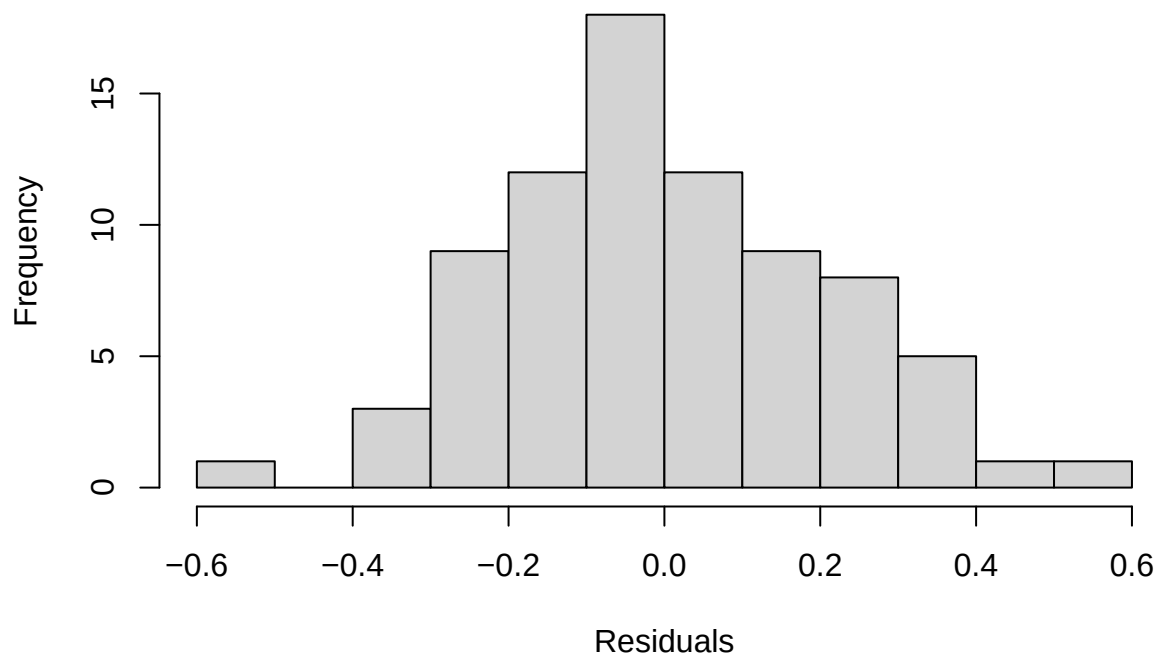
4.1 Change in the total CHC amount over time

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(normalized_mass)) ~ mean_delta + (1 | colony)
## Data: separation_data[-74, ]
##
## REML criterion at convergence: 9.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.55998 -0.69772 -0.07088  0.67280  2.51792
##
## Random effects:
## Groups   Name            Variance Std.Dev.
## colony   (Intercept)  0.01765   0.1329
## Residual                    0.04716   0.2172
## Number of obs: 79, groups: colony, 11
```

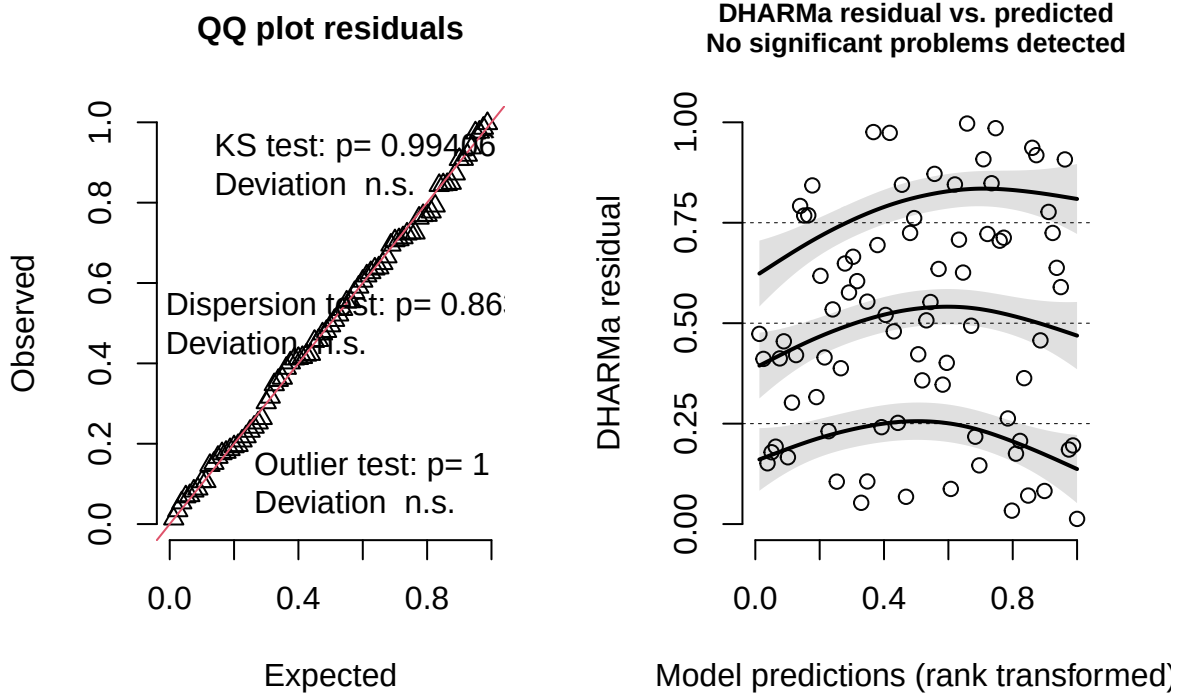
```
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  1.367210   0.055393 16.298104  24.682 2.43e-14 ***
## mean_delta   0.002934   0.001933 68.480649   1.518   0.134
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## mean_delta -0.515

##
## Shapiro-Wilk normality test
##
## data:  residuals(lm_model)
## W = 0.9908, p-value = 0.8482
```

Model conditional residuals



DHARMA residual

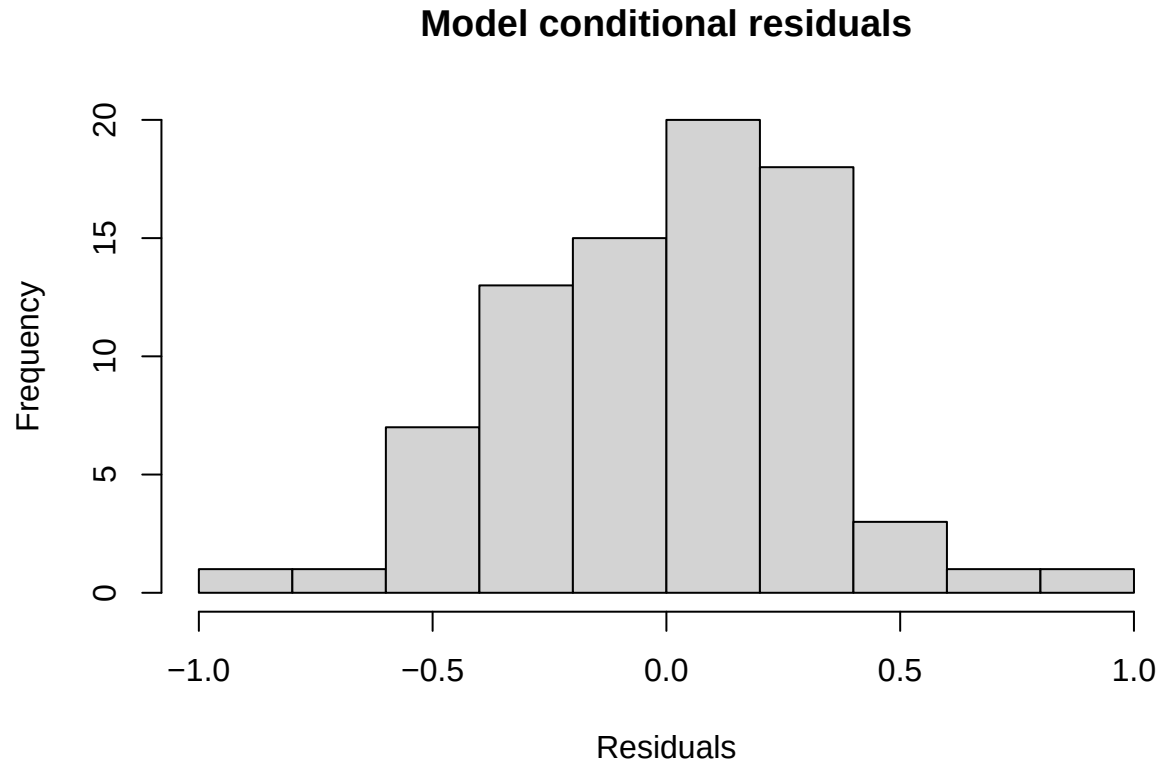


4.2 Change in the amount of compounds characteristic of callow *F. sanguinea*

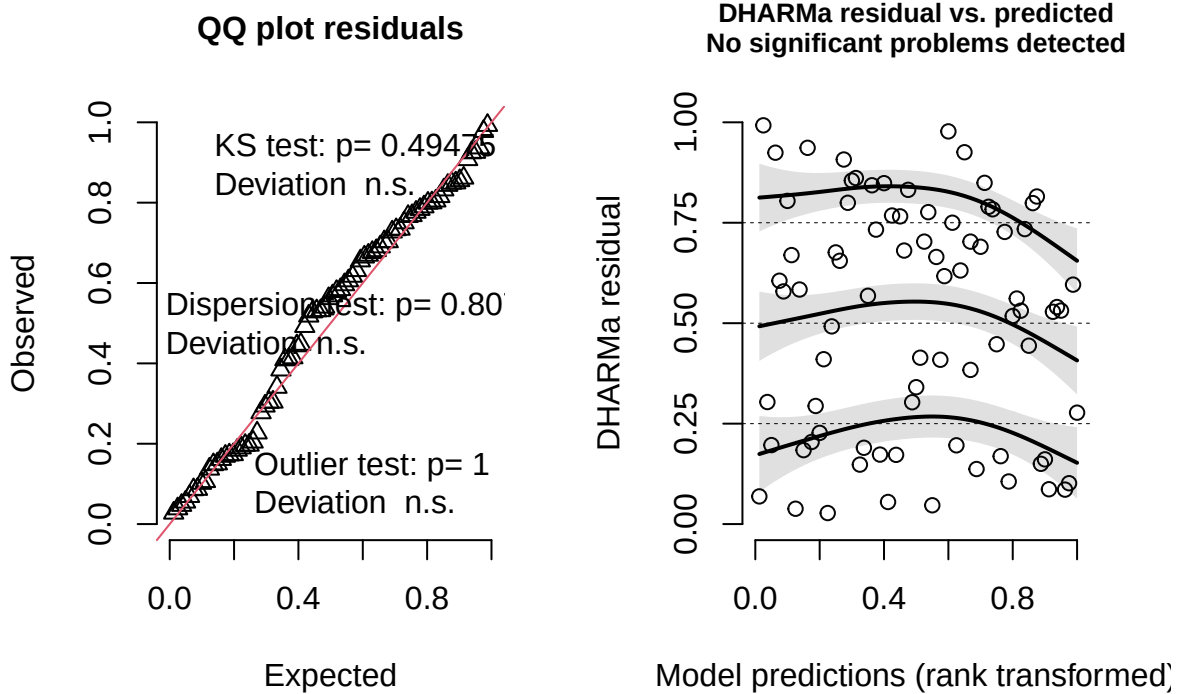
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: log(normalized_mass_part) ~ mean_delta + (1 | colony)
## Data: separation_data
##
## REML criterion at convergence: 92.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.3303 -0.7669  0.1012  0.6201  2.5706
##
## Random effects:
## Groups Name Variance Std.Dev.
## colony (Intercept) 0.1129  0.3360
## Residual          0.1238  0.3518
## Number of obs: 80, groups: colony, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -0.003285   0.118687 13.016273  -0.028   0.978
## mean_delta  -0.004592   0.003054 68.334089  -1.503   0.137
##
## Correlation of Fixed Effects:
##              (Intr)
```

```
## mean_delta -0.389

##
## Shapiro-Wilk normality test
##
## data: residuals(lm_model)
## W = 0.98886, p-value = 0.7216
```



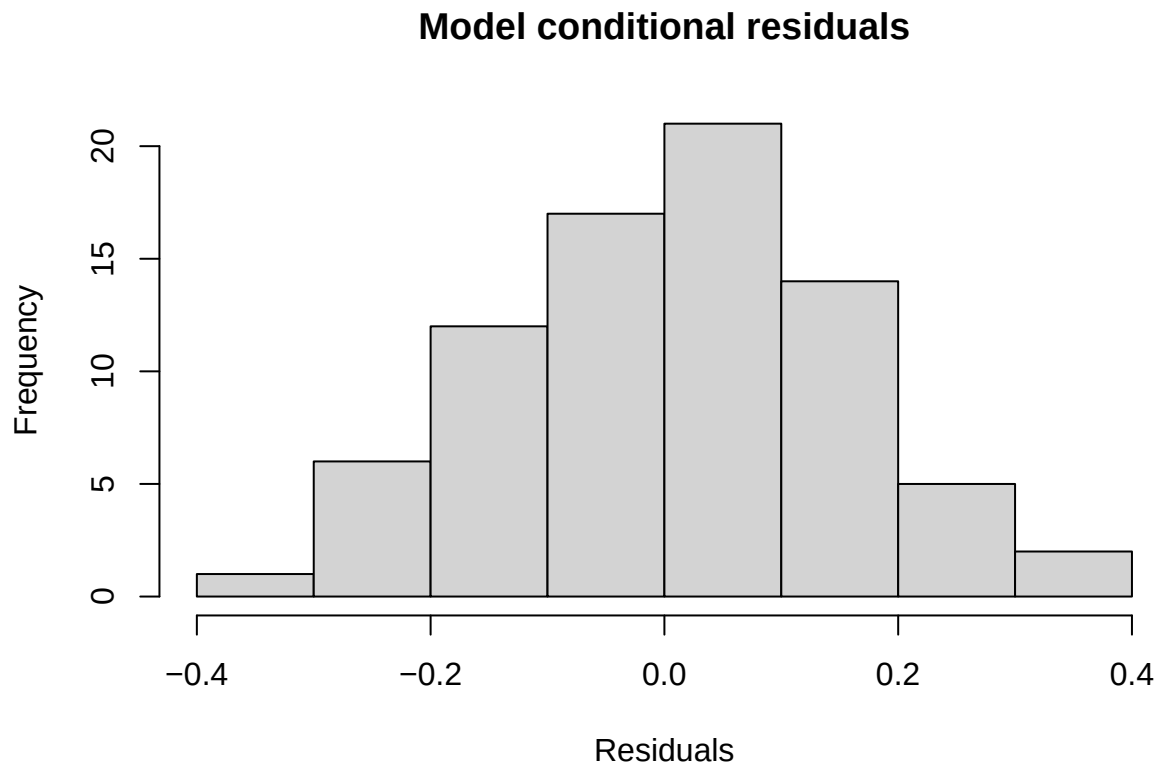
DHARMA residual



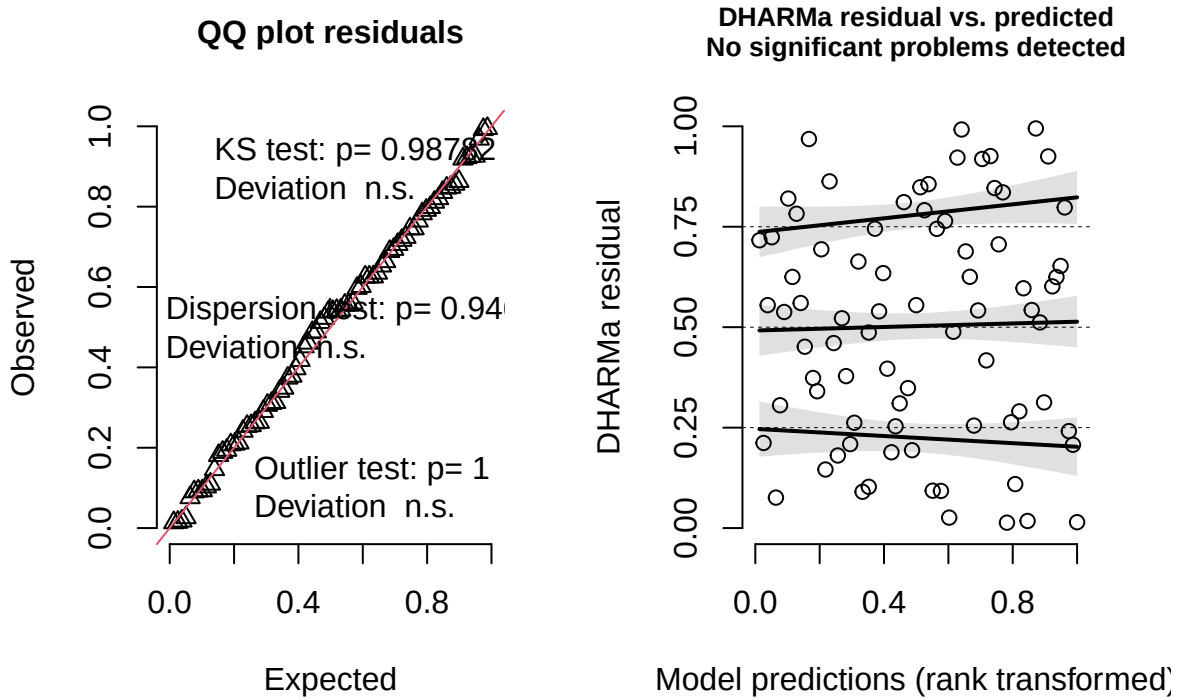
4.3 Change in the amount of compounds characteristic of *F. sanguinea*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: I(sqrt(normalized_mass_part)) ~ poly(mean_delta, 1) + (1 | colony)
## Data: separation_data[-c(69, 74), ]
##
## REML criterion at convergence: -67.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.27083 -0.64402  0.06067  0.73259  2.42696
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## colony   (Intercept)  0.002444  0.04944
## Residual                    0.021031  0.14502
## Number of obs: 78, groups: colony, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.60656    0.02252  7.88363  26.930 4.81e-09 ***
## poly(mean_delta, 1) 1.08605    0.14693 69.21997   7.392 2.54e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:  
##      (Intr)  
## ply(mn_d,1) 0.020  
  
##  
## Shapiro-Wilk normality test  
##  
## data:  residuals(lm_model)  
## W = 0.98936, p-value = 0.7684
```



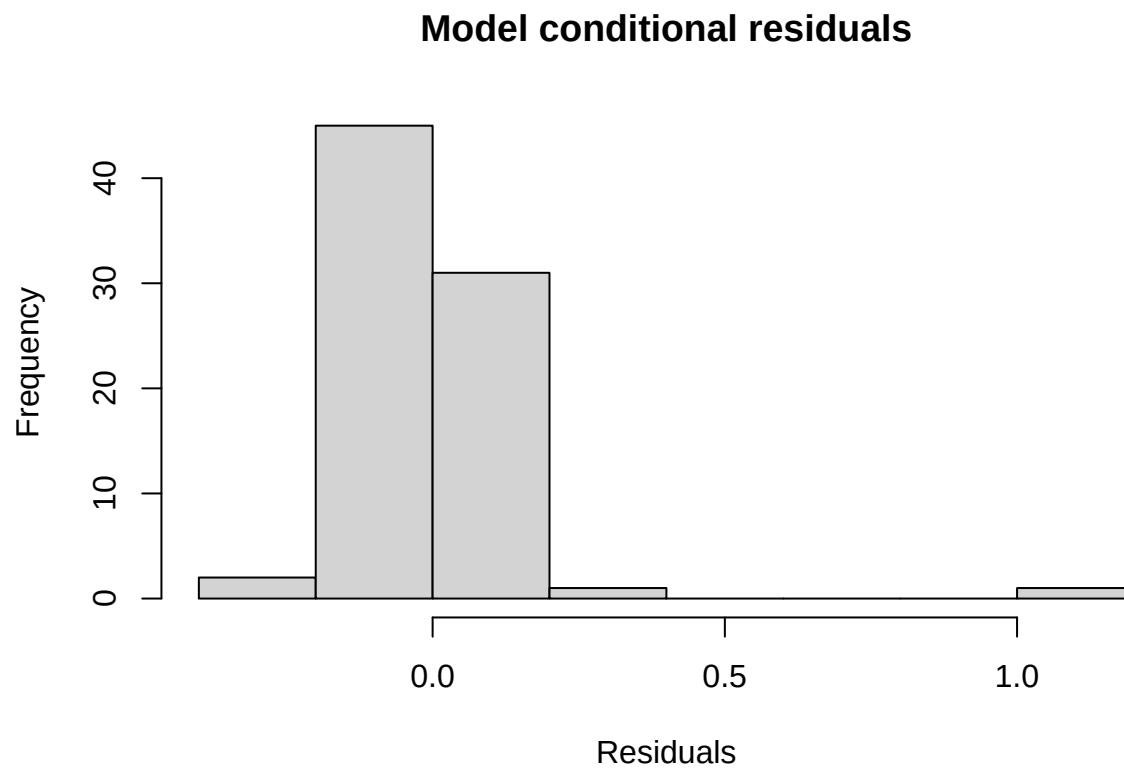
DHARMa residual



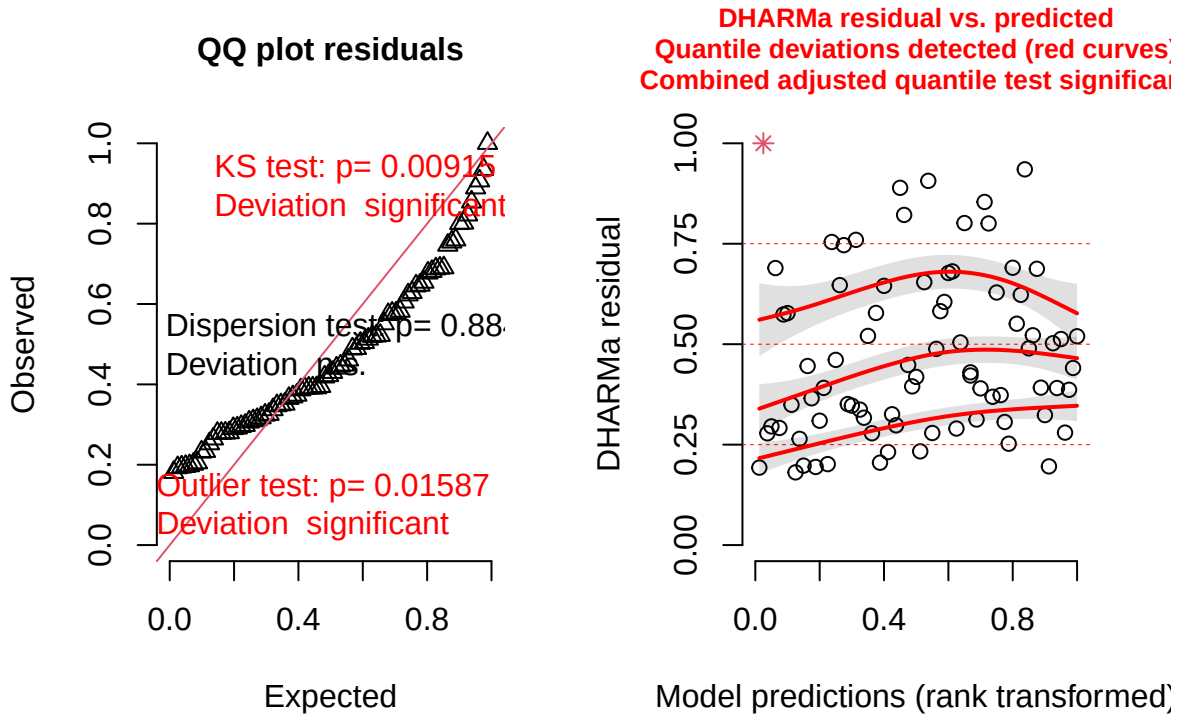
4.4 Change in the amount of compounds characteristic of *F. fusca*

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: normalized_mass_part ~ mean_delta + (1 | colony)
## Data: separation_data
##
## REML criterion at convergence: -25.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.5589 -0.4676 -0.0943  0.2358  6.8831
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   colony   (Intercept) 0.02205  0.1485
##   Residual                0.02765  0.1663
## Number of obs: 80, groups: colony, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  2.240e-01  5.346e-02 1.383e+01   4.19 0.000932 ***
## mean_delta   -4.298e-05  1.443e-03 6.870e+01  -0.03 0.976326
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:  
##      (Intr)  
## mean_delta -0.408  
  
##  
## Shapiro-Wilk normality test  
##  
## data:  residuals(lm_model)  
## W = 0.59858, p-value = 1.852e-13
```



DHARMA residual



5 Impact of slaves on the development of the *F. sanguinea* CHC profile

We analyzed the impact of *F. fusca* slaves on the CHC profile of callow *F. sanguinea* ants. In this experiment, *F. sanguinea* ants were isolated in pairs before eclosion, and their CHC profiles were analyzed at one of four time points marking their adult age. We then tested whether callow *F. sanguinea* ants were chemically more similar to their slave relatives from free-living colonies. Unrelated *F. fusca* ants served as a background group.”

5.1 Analysis based on all samples

```
## Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000
## Number of colonies used in the treatment of 1-3 days of separation: 9
## Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00000
## Number of colonies used in the treatment of 8-10 days of separation: 9
## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00079
## Number of colonies used in the treatment of 17-20 days of separation: 9
```

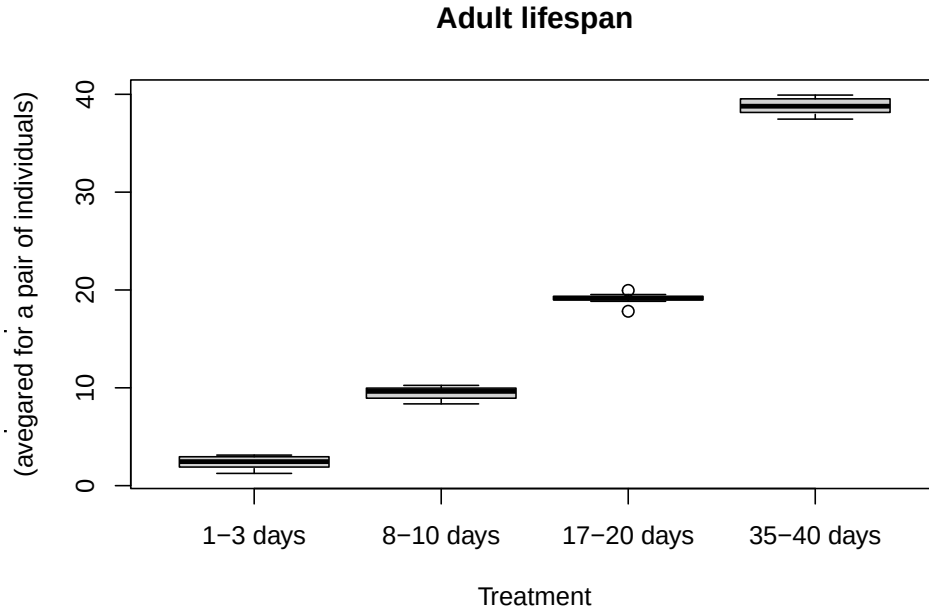


Figure S1 : Distribution of lifespans of ants used in the experiment. The precise age of individuals could not be determined because ants in pairs were not marked. Therefore, all possible combinations were considered to illustrate the potential range.

Table S1 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realted and unrelated to slaves.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3332	0.4718
8-10 days	0.3518	0.4802
17-20 days	0.4521	0.5291
35-40 days	0.5729	0.6100

Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.00703

Number of colonies used in the treatment of 35-40 days of separation: 8

5.2 Analysis restricted to the individuals hatched from cocoons

Since naked pupae might have acquired CHC through physical contact with adult ants after pupation, we subset our data to include only those samples which derived from the ants hatched from cocoons. In this case silky envelop during pupa stage ruled out possibility of the CHC transfer from the environment.

Empirical p-value for the chemical distance difference for the ants aged 1-3 days:0.00000

Number of colonies used in the treatment of 1-3 days of separation: 8

Empirical p-value for the chemical distance difference for the ants aged 8-10 days:0.00001

Table S2 : Chemical distance of the isolated *F. sanguinea* ant to *F. fusca* ants realed and unrelated to slaves. Isolated ant spun a silky envelope before pupal stage, which prevented CHC transfer from the enviroment.

Adult age	Chemical distance to slaves' relatives	Chemical distance to random colonies
1-3 days	0.3473	0.4869
8-10 days	0.3740	0.4932
17-20 days	0.4643	0.5407
35-40 days	0.6175	0.6458

```
## Number of colonies used in the treatment of 8-10 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 17-20 days:0.00438

## Number of colonies used in the treatment of 17-20 days of separation: 8

## Empirical p-value for the chemical distance difference for the ants aged 35-40 days:0.03298

## Number of colonies used in the treatment of 35-40 days of separation: 7
```

6 Effect of dummy ants

In these experiments, *F. sanguinea* ants released from their cocoon envelopes were maintained in Petri dishes along with glass beads coated with CHC extracted from either *F. fusca* or *F. sanguinea* ants. We examined whether the contact with dummy nestmates influenced the development of CHC profile of young *F. sanguinea* workers. In particular, we aimed to determine there there is evidence for active chemical mimicry in *F. sanguinea*.

6.1 Distance to the CHC profile of the dummy ants treated by *F. fusca* CHC

We computed the chemical distance to the CHC profile of ants whose CHC were used to cover glass beads serving as dummy ants. In the control variant, we used the ants subjected to clean glass beads.

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: to_test$mean_dist.x and to_test$mean_dist.y
## V = 18.5, p-value = 0.6781
## alternative hypothesis: true location shift is not equal to 0
```

6.2 Distance to the CHC profile of the dummy ants treated by *F. sanguinea* CHC

```
##
## Wilcoxon signed rank test with continuity correction
##
## data: to_test$mean_dist.x and to_test$mean_dist.y
## V = 12, p-value = 0.2361
## alternative hypothesis: true location shift is not equal to 0
```

Table S3 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. fusca* ants from colonies that served as a source of CHC to coat the glass beads. In control variant, glass bead were left clean.

Colony ID	Treatment	Averaged distance	Contrast	Averaged distance
SD18-3	12-15 days + <i>F. fusca</i> hydrocarbons	0.396	12-15 days (control)	0.509
SD18-5	12-15 days + <i>F. fusca</i> hydrocarbons	0.616	12-15 days (control)	0.513
SD18-5	12-15 days + <i>F. fusca</i> hydrocarbons	0.842	12-15 days (control)	0.832
SD19-11	12-15 days + <i>F. fusca</i> hydrocarbons	0.467	12-15 days (control)	0.469
SD19-3	12-15 days + <i>F. fusca</i> hydrocarbons	0.500	12-15 days (control)	0.436
SD19-4	12-15 days + <i>F. fusca</i> hydrocarbons	0.305	12-15 days (control)	0.424
SD19-6	12-15 days + <i>F. fusca</i> hydrocarbons	0.674	12-15 days (control)	0.699
SD19-8	12-15 days + <i>F. fusca</i> hydrocarbons	0.595	12-15 days (control)	0.587
SD20-2	12-15 days + <i>F. fusca</i> hydrocarbons	0.699	12-15 days (control)	0.709

Table S4 : Chemical distance of separated *F. sanguinea* ants to the CHC profile of *F. sanguinea* ants from colonies that served as a source of CHC to coat the glass beads. In control variant, glass bead were left clean.

Colony ID	Treatment	Averaged distance	Contrast	Averaged distance
SD18-2	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.571	12-15 days (control)	0.373
SD18-3	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.457	12-15 days (control)	0.599
SD18-5	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.525	12-15 days (control)	0.580
SD18-5	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.250	12-15 days (control)	0.232
SD19-11	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.595	12-15 days (control)	0.607
SD19-3	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.419	12-15 days (control)	0.511
SD19-4	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.648	12-15 days (control)	0.668
SD19-6	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.553	12-15 days (control)	0.591
SD19-8	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.850	12-15 days (control)	0.850
SD20-2	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.646	12-15 days (control)	0.654

Table S5 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. fusca* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-3	12-15 days + <i>F. fusca</i> hydrocarbons	0.018	12-15 days (control)	0.000
SD18-5	12-15 days + <i>F. fusca</i> hydrocarbons	0.001	12-15 days (control)	0.000
SD18-5	12-15 days + <i>F. fusca</i> hydrocarbons	0.002	12-15 days (control)	0.000
SD19-11	12-15 days + <i>F. fusca</i> hydrocarbons	0.011	12-15 days (control)	0.002
SD19-3	12-15 days + <i>F. fusca</i> hydrocarbons	0.009	12-15 days (control)	0.001
SD19-4	12-15 days + <i>F. fusca</i> hydrocarbons	0.017	12-15 days (control)	0.000
SD19-6	12-15 days + <i>F. fusca</i> hydrocarbons	0.009	12-15 days (control)	0.001
SD19-8	12-15 days + <i>F. fusca</i> hydrocarbons	0.003	12-15 days (control)	0.002
SD20-2	12-15 days + <i>F. fusca</i> hydrocarbons	0.009	12-15 days (control)	0.005
W17-1	12-15 days + <i>F. fusca</i> hydrocarbons	0.021	12-15 days (control)	0.001

6.3 Fraction of *n*-docosane in the CHC profile

For the control treatment, glass beads were left uncoated with CHC. In the experimental treatments, the CHC used to coat the glass beads were supplemented with *n*-docosane, which served as an internal standard. We examined differences in the proportion of *n*-docosane in the CHC profiles of the tested ants, as these could indicate the acquisition of chemicals from the dummy ants.

6.3.1 *F. fusca*

```
##
## Wilcoxon signed rank test with continuity correction
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 55, p-value = 0.005857
## alternative hypothesis: true location shift is not equal to 0
```

6.3.2 *F. sanguinea*

```
##
## Wilcoxon signed rank test with continuity correction
##
## data:  to_test$C22_prop.x and to_test$C22_prop.y
## V = 54, p-value = 0.007093
## alternative hypothesis: true location shift is not equal to 0
```

Table S6 : Proportion of *n*-docosane in CHC extracted from *F. sanguinea* ants maintained with the glass beads coated with the CHC of *F. sanguinea* ants and contaminated with *n*-docosane. In control variant, glass bead were left clean.

Colony ID	Treamtent	Averaged C22 fraction	Contrast	Averaged C22 fraction
SD18-2	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.005	12-15 days (control)	0.002
SD18-3	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.014	12-15 days (control)	0.000

SD18-5	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.003	12-15 days (control)	0.000
SD18-5	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.002	12-15 days (control)	0.000
SD19-11	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.004	12-15 days (control)	0.002
SD19-3	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.003	12-15 days (control)	0.001
SD19-4	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.002	12-15 days (control)	0.000
SD19-6	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.004	12-15 days (control)	0.001
SD19-8	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.002	12-15 days (control)	0.002
SD20-2	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.004	12-15 days (control)	0.005
W17-1	12-15 days + <i>F. sanguinea</i> hydrocarbons	0.003	12-15 days (control)	0.001

7 Body surface area of *F. sanguinea* ants and their slaves

In the publication, CHC amounts are reported relative to the square of head width. To assess whether this measure reliably reflects body size, we measured the planar projection areas of different body parts of *F. sanguinea* and *F. fusca*. The plots below suggest that head width slightly underestimates body surface area in *F. fusca* compared to *F. sanguinea*. As a result, the relative concentration of CHC in *F. fusca* may be overestimated. However, this bias acts in the opposite direction of our findings—*F. sanguinea* exhibits greater CHC concentration on its cuticle. Therefore, the shape-related bias between species is conservative, as it reduces the likelihood of rejecting the null hypothesis

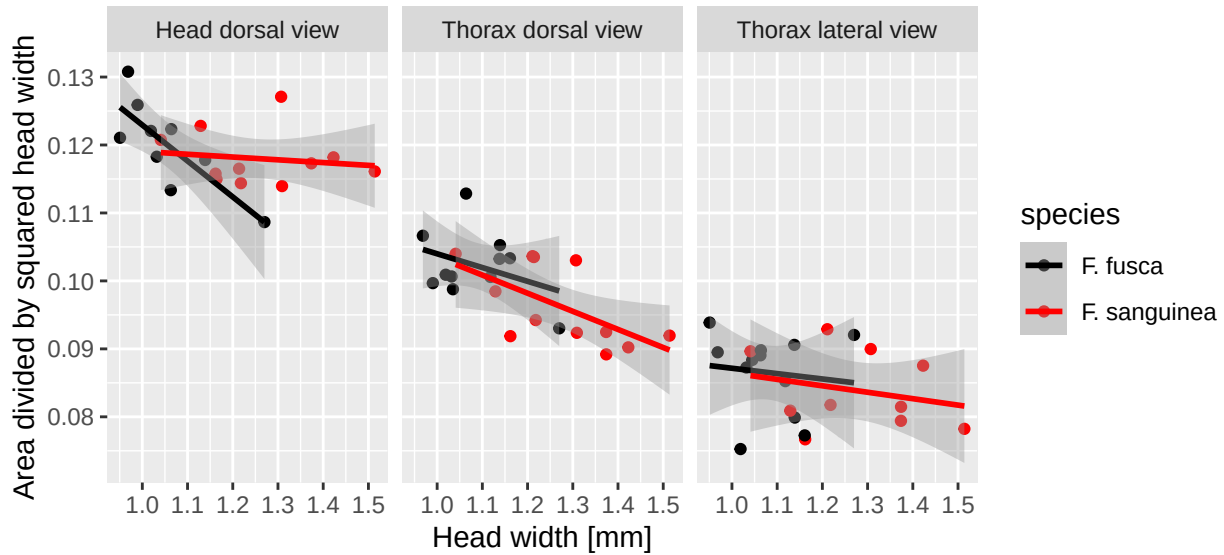


Figure S2 : Ratios of the planar projection areas of three body parts (head, thorax dorsal view, thorax lateral view) to the square of head width.

References

Florian Hartig. *DHARMA: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*, 2024. URL <http://florianhartig.github.io/DHARMA/>. R package version 0.4.7.